Cedric (Physio Student)

Interviewee Consent and Profile

The following questions are to build a profile of the interviewee in their relevance to the development of the device, as well as confirm their experience with stroke patients and/or movement specialisation.

Name	Emmanuel Cedric Olivas
Occupation (Physiotherapist, Kinesiologist, or Physiatrists, etc.)	
Date	05/07/25
Our Introduction	We are a group of final-year students from the National University of Singapore (NUS), currently working on a research project as part of the CDE4301 Final Year Project module. Our project focuses on the viability and development of a robot-assisted rehabilitation device targeted at stroke patients' unilateral lower limb/extremity (LE) to be used in supine or sitting exercises. The aim of our device is to support the rehabilitation process by improving patient outcomes while also reducing the physical workload on physiotherapists during therapy sessions. We are currently gathering professional insights to guide the design, usability, and practical relevance of our solution.
Scope of Interview	Thank you for agreeing to participate in this interview. The purpose of this interview is to gather expert insights that will inform the development of a robot-assisted rehabilitation device for the lower limb aimed at supporting stroke patient recovery. What your participation involves: ● A short, recorded, and noted, interview lasting approximately

	30-40 minutes. ● Questions related to your professional experience with stroke patients, particularly in movement rehabilitation. ● No personal or identifying information about patients will be requested. Your rights: ● Your participation is completely voluntary. ● You may decline to answer any question or stop the interview at any time. ● Your identity will be made anonymous if requested. The information you provide will be used solely for academic or research purposes.	
Consent Statement	I understand the purpose of this interview and consent to participate.	Y/N
	I agree that the information I provide can be used anonymously for research and documentation related to this project.	Y/N
	I agree to be recorded for transcription purposes.	Y/N
Verification of Patients Experience	What is your experience with stroke patients in terms of years or number of patients?	What settings have you worked in (acute, rehab, or outpatient)?
	Elderly is more common to see, not rly young adults	

Questions (To be organised)

Question	Rationale (What this question can help with)	Response
Patient Evaluation		
What is your typical initial physical assessment of stroke patients, and what are the criteria you use to determine treatment readiness?	**UI/EE/ME** Understand our demographic	Try to get to active training as quickly as possible (post 24 hr treatment)
What measures do you use to track progress?	**UI/EE/ME** For deciding which data to collect/sensors to use	Bodyweight > Smaller additional weights > Slowly increase weights More concerned with functional activities that can be done - Walking test - Endurance (walk for 6 min and measure distance covered depending on age grp) - Gait speed (how fast can they walk, .74m per second for elderly community speed) - How fast can you get up and down from the chair - Last sig one is fall risk

		management, aka single leg stance, - Every hospital has their own standard of measures Exact force produced is not important, quantifying is not a focus, just enough for daily needs
What are some ever used mechanical or robotic assistance during physiotherapy on the lower limb (LE)? Please elaborate on the types of devices you have employed: (Try to get their purposes and the challenges faced)	**UI/EE/ME** When is it that physiotherapists choose to employ devices so that we know what pain points we are addressing	Current landscape is not enough bc of lack of manpower, stroke patients the more the merrier.
What kind of exercises are performed on the LE in supine and sitting position?	**UI/EE/ME** For us to understand better about the degree of motion of our device	Passive exercises - secondary treatment - to avoid worsening of condition, not restorative Active exercises are more relevant to treatment, with focus on preventing tissue contractures. Grade 1/grade 0 is when we use functional electrical

		stimulation, and then taking off some of the weight (Manual Muscle Grading - Physio Standard Test) The muscle itself is not super weak, we just don't have access to it. So determining amount of assistance is not important, the regaining coordination is important, but just enough so that the movement can be completed, followed by normal resistance training
Compensatory movements can hinder recovery. What are some common LE compensatory movements that occur and what do you do to minimize them? (Using the hips/momentum instead of executing controlled movements)	**UI/EE/ME** Planning how our device can minimise 'wrong' motions so that it is effective, or how the UI can give warnings	Video is accurate. The physio will use visual cues. But for gait training, a bit of compensation is okay. Orthosis can be used to restrict certain parts to make sure patients use the right muscles, or use splints or tape on the non-paretic side
How do you typically communicate recovery to patients? OR what is the feedback that patients need from you during a session?	**UI** How should the UI communicate progress after each session?	Biggest difficulty of the communication is empathy > Will need to provide a "reward" for elderly stroke patients to encourage them

		and improve self-esteem > Difficulty adjustment for the device? Language difficulty (Cannot understand what you are saying) > Simple designs and gestures on the UI? UI can have different modes for communication (reading, audio, visuals, or using simple commands, etc)
What are some difficulties you face when conducting physiotherapy sessions? (Large number of patients, physically taxing, etc)	**UI/EE/ME** Pain points again	Main difficulty - Lack of manpower Treatment - The more exercises the patient can do, the better
Do you think there is a lack of equipment specialised for stroke patient rehabilitation? And do you believe that this device can assist their rehabilitation? If not, why do you think so?	**UI/EE/ME** Justification for research and development for device	Devices can definitely help rehab because of manpower issues. Robot-assisted gait training and treadmill (bodyweight-supported)
Physiotherapy Session		
What is the frequency/length of sessions, and how do you determine the duration of one rep of exercise?		Acute stroke (everyday of the week). Rehab: 1 hour to 90 minutes / session (do up to 3 months). Expected to recover the most function in the first 3 months Speed of rep differs patient to

		patient due to spasticity, too fast can cause ‘catch in muscle’, speed is not prioritised, slower/poorly coordinated is expected. Standard procedure does not really exist, reccs no set speed, physiotherapists want patients to be able to the exercise as much as possible (“a good number” doesn’t exist) → goal is to complete functional movements asap that may differ from patient to patient (1000 reps accomplished?)
How to design a treatment plan?		Directly do as many reps as possible. After discharged: can come for regular visits (depends on the severity)
Where is the equipment stored?		Depends on the hospital
Are elderly patients lying down in bed during rehab sessions? Or are they seated?	**ME** To determine the device design (mechanism)	They sit down a lot. Recommended for patients to sit in an inclined position on bed. Help their respiratory function as well.
What are your thoughts on the feasibility of KINETIX in different contexts? (E.g. Hospice settings, etc.)		The most important use is in community hospitals. Stroke patients usually usually go to community centers instead of hospice.
These are the metrics that we aim to communicate to physiotherapists to monitor progressive recovery. What are your thoughts on their usefulness?		Daily fatigue scale and RPE are used to read the patient Angular velocity is not really important for the physio. Other than that, other variables are important

Range of Motion (ROM), Joint Angles Angular Velocity & Smoothness Force / Strength Estimation (Normalised to body weight) Repetition Counting & Segmentation Trajectory Tracking Muscle Fatigue Estimation Consistency of Movement		
Which method of disinfection is preferred by physiotherapists (disposable paper or sanitizer, etc)?		Bed sheets would be swapped regularly. Alcohol wipe would be used for sanitise. (hospicare alcohol wipes) Only worry: Device has to have comfort as the priority (elderly skin is extremely frail) due to infection risk that might be caused. Avoid chafing
When using larger tools/devices for physiotherapy, what is the setup and storage process like?		Setup shld be 10 minutes max (turn on, calibrate, etc), robots make them feel less independent sometimes even tho it looks cool
Before starting a session, what kind of preparatory questions do you ask? For example: 1. Any joint pain? 2. Have you gotten enough sleep recently? 3. Mental state?		1. Any pain? 2. Rapport building? 3. How well are they performing their task? → usually asked in between weeks to months

Other Notes:
For passive exercises, we just mainly try to avoid tissue contracture, which affects ROM. For standard int, we recc a lot of active exercises as well.
Generally agrees that device has value in reducing burden

Better for them to do exercises as much as possible, severe cases need to get out of bed as soon as possible and walk of their own, 2-3 physios is difficult to spare, always an influx of patients tldr

Add restraint for the patient's (partially passive; future feature). Fully passive exercise is a secondary treatment (preventative measure). Secondary exercise is used to avoid muscle stiffness (tissue contractor).

Sensory intervention(?), stroke patients don't have sensation on the side of hemiplegia, so machine usage here is something to consider, we want more patient-led treatments bc physios and patients may have different expectations, and consider the mood a lot, loss of autono, don't say it'll be okay

Btw can't go fast with exercises, fail safe, rom must be

Fail and Safe mechanisms should exist (safety measure in case there's something wrong with the machine). Range Of Motion design (adjustable).

First 3 months high reps to rewire the brain function to the lower limb

Robotics could enhance the reps on the patient.

Disclaimer: This interview transcript has been lightly edited for brevity and clarity. Filler words, repetitions, and unrelated exchanges have been removed, but the meaning, tone, and conversational style have been preserved.

Part 1

[START]

00:13:36 Sebastian

I think you're muted. OK, alright. All is good. So welcome. Cedric. These are my team members. Oh, yeah, it it's not too formal. You could be chill

00:13:46 Sebastian

However you like.

00:13:46 Cedric

OK. OK. OK.

00:13:47 Sebastian

It yeah, there's no teachers here. There's no problem. It's just us students.

00:13:51 Cedric

OK. Sure. Hello. Hi. Hi. Hi.

00:13:52 Sebastian

All right, so this, yeah, this is not. Yeah, Rizqi. Hello. Hi.

00:13:58 Sebastian

So Ricky is our robotics expert because yeah. So he has a lot of experience with it and we have Nadia here who is taking a minor data

engineering or computing?

00:14:09 Cedric

Hmm.

00:14:11 Sebastian

OK, alright, computing OK, so just a very quick introduction to what we're doing. So you're not totally lost. So for our final year project, we're basically focusing on robot assisted rehab devices. Mostly focused towards stroke patients like they're starting to recover on like the lower half. So maybe like their legs.

00:14:18 Cedric

MHM.

00:14:32 Sebastian

The thighs, that part of.

00:14:35 Sebastian

Their body. And so this like robot is like passive movement. So basically they're just gonna lie down on it. Or or are they sitting down on their bed and this device will move their legs for them if.

00:14:48 SEBASTIAN

You catch my drift.

00:14:49 Cedric

Hmm.

00:14:50 Sebastian

OK. So we're just trying to support the rehab process. So by actually just reducing the physical workload.

00:14:58 Sebastian

Uh on the for you guys as well because you're going to be a future physio, right?

00:15:01 Cedric

Yep, Yep.

00:15:02 Sebastian

Yeah. So it's, they're just trying to help you all out and we just want to see like whether from your perspective, is it like useful and if you could also help us with your expert advice, because we're not really familiar with this.

00:15:14 Cedric

Expert advice, OK.

00:15:15 Sebastian

Yeah.

00:15:16 Sebastian

I mean, there's a reason why you're helping us on the basketball whenever we get injured.

00:15:19 Cedric

Yeah.

00:15:20 Cedric

OK. Yeah, OK. But just just note, I'm you're physio, right? So I mean, my advice is as far as what I've learned so far, but yeah, OK, so.

00:15:22 Sebastian

OK.

00:15:32 Cedric

Right now I'm already aware that there's a lot of robot assisted.

00:15:36 Cedric

Training in stroke. So actually it's quite popular even in Singapore, although that's only in the hospital wards they it's not. You can't take it home. So it's only for inpatient stroke patients. So right now we have the robot assisted gait training.

00:15:53 Cedric

So it's literally just helps the patient walk.

00:15:56 Cedric

So in that regard, like for your robotic assistance, like what?

00:16:02 Cedric

Is like what? What was the differentiation versus this RAGT? The robot assisted gait training.

00:16:11 Sebastian

OK, so for that part.

00:16:14 Sebastian

We have. Do you have the link of the video, Ricky, maybe you can share like the gate thing. Yeah, like the one that they did. So using that as reference for our like like what exact movements we are supposed to be doing. I think you have that video, right I.

00:16:20 Rizqi

Right.

00:16:28 Rizqi

Oh, sure, OK.

00:16:31 Sebastian

Don't. I don't have it with me.

00:16:31 Rizqi

Oh I.

00:16:32 Rizqi

I just share the link or show screen.

00:16:34 Sebastian

Yeah, just should just.

00:16:35 Sebastian

Take over my screen or something.

00:16:36 Rizqi

Oh, sure, sure. Let me just open my YouTube very quick.

00:16:43 Rizqi

And then share the screen with you guys.

00:16:46 Rizqi

Hold up.

00:16:57 Rizqi

OK, just share my screen.

00:17:07 Rizqi

OK. So can you see the the video, can you see the YouTube video? Sorry, the YouTube. OK, so yeah. So for my design I used this video as uh, my

main reference. So this one is from the Singapore General Hospital like it's the.

00:17:26 Rizqi

Exercise done by physiotherapists to towards the post-stroke lower limb patients.

00:17:34 Rizqi

So let me just make it a bit slower, 1.5.

00:17:51 Youtube Video

The patient had just started.

00:17:56 Youtube Video

Next to the same handhold.

00:17:58 Youtube Video

Bring the leg away from the.

00:17:59 Youtube Video

Other leg hold for 20 seconds.

00:18:03 Youtube Video

Next, bend the knee and need to 90° fix the knee and rotate the leg in oh for 20 seconds we'll take the leg up, hold for 20 seconds.

00:18:20 Youtube Video

Resistance.

00:18:27 Youtube Video

For the ankle support under the heel in the position shown, with the other hand, keep the knee straight using your body weight. Stress the

angle by bring the foot upwards towards the head. Ohh for 20 seconds.

00:18:39 Rizqi

OK. So yeah, that's so I used this video as my reference to for my my design of the of the of the device. Maybe I just share a bit on how does it. How would it look like. So here's my sketch of the device itself. So.

00:18:58 Rizqi

Right now it's still within prototyping stage. It's a bit scuff, but I hope you get the idea here. So later the like of the patient will be strapped towards these this.

00:19:13 Rizqi

Linkage linkage system. So we have several.

00:19:20 Rizqi

Several movement for for OK for this one is for, for aiming at towards the the knee muscle.

00:19:29 Rizqi

And then from here you can also move side to side.

00:19:33 Rizqi

Yes, we'll do this.

00:19:40 Rizqi

Hold up, OK?

00:19:43 Rizqi

OK, so also it can move side to side.

00:19:49 Rizqi

Kind of. OK, right. Always say, OK, I mean, I mean.

00:19:54 Rizqi

Ah.

00:19:56 Rizqi

Sorry guys.

00:19:58 Rizqi

So let me just move this.

00:20:02 Rizqi

OK.

00:20:05 Rizqi

OK. And also able to move side to side to target that muscle joints also.

00:20:14 Rizqi

Able to move up.

00:20:20 Rizqi

Like move up and down. So later these movement would be would be controlled by a motor to be be to be able more automated. So if I put the sketch right, I actually have the sketch right here to to give a better, a better.

00:20:40 Rizqi

Picture.

00:20:41 Rizqi

So these are the different modes that I, we we planned to to achieve.

So basically there are one 2-3 different modes and you can see right here the on the joints like on the like I I pointed out which muscle or joints that I want to be to targeted to be stimulated.

00:21:01 Rizqi

For the exercise.

00:21:03 Rizqi

Yeah. So this is the knee. And then and then we here and then over here. OK, I think that's it.

00:21:14 Rizqi

Yes.

00:21:16 Nadia

I think just summarize uh with the robot assisted day training. Uh, there are lot of devices these days which focus a lot more on gait training and ambulation.

00:21:26 Nadia

Where is the devices that are more for supine and seated exercises, is what we want to focus on.

00:21:32 Nadia

Especially now, since there's a whole, you know, shortage of healthcare workers and professionals. So if we want to see how far we can, uh, make a device that can be independent and also take off some of the burden.

00:21:43 Nadia

For the physiotherapist.

00:21:47 Cedric

Hmm. OK, so actually, yeah, I think definitely if you match to implement this I I think it it will assist a lot in the healthcare wards.

00:21:58 Cedric

Because uh, so the exercises prescribed by the SGH. Those typically are so like when we do treatment for stroke patients, the whole focus is obviously it's patient recovery, right? But though.

00:22:17 Cedric

For the passive exercises such as these, the main intervention that we're looking at here is just to avoid tissue contracture, which means is like the muscle becomes stiff and it becomes shortened, so then they lose their range of motion.

00:22:31 Cedric

So when you give provide these passive exercises, it will assist the lot for the patient comfort.

00:22:38 Cedric

But I do have to say that.

00:22:41 Cedric

Or standard interventions now, like the treatment is we do recommend the patient to do active exercise as much as possible. But if you guys like pivot towards net actively aiming towards avoiding the contracture and shortening of the tissues in stroke patients, I think this will definitely help.

00:23:01 Cedric

The overall burden for healthcare workers.

00:23:06 Sebastian

Do you think?

00:23:08 Sebastian

Because right now for my understanding of stroke patients, they're just different severities, right. So.

00:23:13 Cedric

Yeah, definitely.

00:23:16 Sebastian

We're still trying to decide whether to aim for mostly the elderly, which in this case this would be more applicable or look or like young adults then active movement would be much more better in that sense. So I'm not sure. Wait they did you have, like, clinicals, as your internship like for physio like you guys have internship.

00:23:36 Cedric

Well, not yet.

00:23:38 Cedric

It’s next year.

00:23:39 Sebastian

Oh, it's next year, but have you met, like, professional physiotherapist?

00:23:45 Cedric

Yeah, of course. Everytime.

00:23:47 Sebastian

Then in do you know like what is their most common group for stroke patients that if you’ve met recently?

00:23:57 Cedric

Definitely elderly, young adults not so much. It's like very rare to be very like, yeah, yeah, yeah, yeah.

00:24:02 Sebastian

Ah, great, it matches with our. OK, good.

00:24:07 Sebastian

All right then.

00:24:08 Sebastian

Just.

00:24:09 Sebastian

In that sense, um, would you? What are some like other exercises would you recommend?

00:24:16 Sebastian

In in terms of like passive rehab exercises for the elderly, like apart from what we should do.

00:24:25 Cedric

Passive exercises.

00:24:29 Cedric

OK, let me think.

00:24:33 Cedric

Hmm.

00:24:34 Cedric

That's a good question.

00:24:39 Cedric

Not really much other than those for. If you're talking about purely passive.

00:24:43 Cedric

Mm-hmm.

00:24:44 Cedric

No, not much, because everything within stroke rehab is really active.

00:24:50 Cedric

All active exercises just the standard intervention like even 17 hours post stroke.

00:24:58 Cedric

Like within 24 hours post stroke they already tried to get the patient out of bed into active movement.

00:25:05 Cedric

So passive exercises are like.

00:25:08 Cedric

It's like more like a secondary treatment as opposed to the primary treatment for stroke patients, yeah.

00:25:14 Sebastian

So in that sense, would you say?

00:25:18 Sebastian

It would be better if we have like a device, specifically focus on, you

know the.

00:25:26 Sebastian

Preventing the muscle contraction part, the one that you talked about because since there's different severities to stroke right, for those, I guess what we can like proposes that have this particular product for really severe cases where they're unable to move like the lower half, right, and then we can transition to a more.

00:25:46 Sebastian

Active movement in a sense.

00:25:49 Cedric

Yeah, I think if you guys aim towards avoiding the contractures part, I think it would be better. But if you want another way, it's more like.

00:25:59 Cedric

You like if you if you want, like not my advice, but like what I think you're like. Robots assisted care can help with is more like active assisted. So like actually like everything that you're doing. You can still like whatever robots you have already or like whatever motor machine you have. I think the movement itself is already OK.

00:26:20 Cedric

But you don't need to make it completely passive for the patient.

00:26:25 Cedric

So like for example like you could just take off maybe 80% of the weight of the leg and then get the get the patient to do it themselves so that at least it's slightly. It's still making the patient exercise.

00:26:37 Cedric

Little bit, yes, we for most patients right, for more severe cases most

of them actually have what we call it's like grade one or grade 0 MMT. So that means they can't even activate the muscle properly.

00:26:53 Cedric

Or they can't even lift up their own leg, so they can't even do body weight exercises like a simple need extension. They're are they able to do that? So right now what we have for them is this thing called functional electrical stimulation, which is like we we just shock their muscles into movie in but.

00:27:12 Cedric

Uh, in reverse what physios do is will help assist them manually by lifting their legs. So we're taking off some of the weight and then getting them to exercise.

00:27:23 Cedric

But if you use the robots instead to to help, like lighten the weight of their own body parts, I think then active assistant can still help a lot.

00:27:33 Rizqi

OK. Yeah, I I think I think we we haven't mentioned that actually that's one of the feature that we also we want to implement that. So the first feature would be is fully fully.

00:27:47 Rizqi

Fully, fully helped by the robot itself. And then the second mode, the the patient would would would move their own limbs by themselves, but like actually can control the motors to add some resistance towards it. So. So yeah as resistance constraints.

00:28:06 Rizqi

So the second mode would be more let's say like.

00:28:12 Rizqi

More guided, not more guided like is yeah, basically. So I think I think, yeah. Yeah. Like you mentioned before the the the supposed to be like some resistance or weight towards the patient or the patient should do the.

00:28:32 Rizqi

Exercise by themselves. Yeah, actually, yeah. That's one of one of the plans also for that. One of the feature that we also want to implement towards our design in the 1st place, yeah, we haven't.

00:28:44 Rizqi

Mentioned that.

00:28:45 Cedric

Yeah. Yeah, yeah, yeah, that is that that one is more important for standard rehab in Singapore.

00:28:54 Rizqi

OK, yeah. As opposed to the completely passive one, that one is still useful, definitely it will definitely help a lot, but it's not. It's like more it's really secondary treatment.

00:29:05 Rizqi

OK, yeah, fully fully passive is small then secondary treatment. OK, OK.

00:29:12 Nadia

Since you say?

00:29:13 Speaker 2

Secondary treatment. Do you mean that physiotherapists will have?

00:29:17 Nadia

Like.

00:29:18 Nadia

Uh. More fixed proper sessions for active training and then the passive movements they're just like.

00:29:27 Nadia

So stretches is it?

00:29:30 Cedric

Yeah. So for usually they'll stroke patients for the severe cases. They'll be inpatient, right? So that means they'll be warded inside the hospital. So when the physios come through their rounds through for each patient, the main like when they sit down the main treatment that they would do.

00:29:48 Cedric

Is to get them out of bed and do the exercises actively, like so they'll assist by taking off the weight of their legs and making them stand up or helping them walk around. Especially especially walk around. But when I mean my secondary treatment is it's.

00:30:06 Cedric

To avoid like what is it called? Comorbidities? Like any such the stroke is the first issue. Obviously the is the first issue that comes apart, but the second issue is the contractures, which is the shortening of the muscle and the stiffening of the muscle and tissue in the leg. So what I mean by secondary treatment is.

00:30:28 Cedric

When you do the passive treat.

00:30:30 Cedric

Passive exercises. It's to avoid the patient getting.

00:30:36 Cedric

Like their condition getting worse as opposed to improving them, yeah.

00:30:40 Rizqi

Oh, it's more like prevention is it.

00:30:43 Cedric

Yeah, yeah, yeah, it's a.

00:30:44 Cedric

Preventative.

00:30:45 Cedric

As opposed to restorative?

00:30:51 Nadia

Then the other.

00:30:51 Nadia

Question I had was when you do active exercises, how do you determine how much assistance to provide? Like how much weight?

00:30:59 Nadia

Do you take off? Yeah.

00:31:00 Cedric

OK, that's quite a good question because as far as I know.

00:31:06 Cedric

It's quite arbitrary.

00:31:08 Cedric

There's no set amount.

00:31:10 Cedric

Because as long as you can get the patient to attempt so the main point of the exercises isn't that.

00:31:18 Cedric

They want them to use their muscles and then get stronger because actually in stroke patients the whole point is that they're unable to.

00:31:26 Cedric

Recruit.

00:31:27 Cedric

Their muscles properly, so it's not like the muscle itself has become super weak because it's different from like when you injure the muscle then it atrophies. This one is just you get stroke and then you you're unable to signal the muscles properly. So how much you want to apply is not as important because you just want them to.

00:31:48 Cedric

Initially to just get them back into trying to coordinate into using their muscles again. Then after that it's great for gaining strength over time. So the start where the physio is providing assistance, that one is I can't say for sure I can't give you a number, but then afterwards it's.

00:32:07 Cedric

They'll start with body weight. Then they'll progressively add more weight and weight like a normal person already.

00:32:13 Nadia

They'll start with body weight. What was that?

00:32:16 Cedric

So the treatment usually is.

00:32:19 Cedric

For a lot of stroke patients, they will be unable to handle their body weight, so then the physio will just provide.

00:32:26 Cedric

Just enough amount that he could start moving the muscle in their in the physiological reject, for example, like if I wanna lift my shoulder like to the side, right.

00:32:36 Cedric

Then, if he's unable to do that, the patient, then I'll just, I just keep applying a little force until he can actually move it in that range.

00:32:44 Cedric

So how much is specific to each patient just so that they can do the movement?

00:32:51 Cedric

OK. Yeah. Then afterwards it's just normal resistance training from there.

00:32:55 Sebastian

So umm would you say that the best way to track the progress of the patient is through rather how much?

00:33:02 Sebastian

Additional resistance that they can overcome, so it's very similar to muscle building, like how you do your typical gym workouts in a sense, right? So it's like increased from body weight and then you have additional resistance.

00:33:09 Cedric

Yeah.

00:33:15 Cedric

For physios we that is true that we have an outcome, this one is called manual muscle testing. So this is like our gold standard for measuring how strong the muscle is. So it's just like.

00:33:26 Cedric

We have grades zero to five, so you want me to run through like each grade. OK, so basically when they're grade 5, that means they're perfectly healthy. But usually, if they're already 3 or 4 means they're already not bad, you can discharge them. So 5 just means perfectly strong. They are able to resist.

00:33:46 Cedric

Uh, they're able to manually resist against an external force applying on that muscle 4 is they can move it within the normal physiological range.

00:33:58 Cedric

Thereby meaning like if I want to raise my arm up, they're able to do so, but they cannot take any extra resistance.

00:34:05 Cedric

Grade 3 is they're able, they're unable to do it within the full range.

So like if I raise my arm up to the side, I can maybe only like lift it up a bit, but I can't bring it up all the way.

00:34:16 Cedric

Then Grade 2 is they're unable to do it against gravity, so you have to do. They're not even strong enough to lift it up.

00:34:24 Cedric

Against gravities like they can't even lift their own body weight, but they can still like. For example if you.

00:34:30 Cedric

Bring your shoulders up while lying down. They can do that. They Grade 1 is they cannot do that even while lying down. Grade 0 is there's no muscle activation at all. OK. Yeah. So that's our standard for, like, measuring how much the person is progressing for each muscle group.

00:34:47 Sebastian

OK. That's really important then, alright.

00:34:50 Cedric

Yeah, but if you're.

00:34:51 Cedric

Talking about like my outcome measures for the patient itself, we're less concerned with that and more concerned with like the functional activities that he can do.

00:35:00 Cedric

Yeah, yeah.

00:35:02 Sebastian

OK, so this question is a bit more. Uh, focus on the actual industry like so I think although it's not really applicable because you're still.

00:35:12 Sebastian

Student and for all students. But I think you have conversed with like actual physiotherapist, right? So maybe you can, like, pass down what they've told you. So in terms of, like, physiotherapy sessions with stroke patients, right. What are some difficulties that do they face? And do you think that right now in Singapore that's like some?

00:35:32 Sebastian

A lack of equipment or resources that specialize for stroke patient. We have. Or do you think that it's already enough?

00:35:44 Cedric

No, it's definitely not enough. So I think the main difficulty for stroke stroke patient rehabs, right is really the lack of manpower.

00:35:53 Cedric

So the reason being this is actually for stroke patients.

00:36:00 Cedric

I don't know how else to say, but basically for treatment the more the merrier. So it's like the more you can get the patient to do the exercises every single day throughout the day, the faster and better the recovery will be. But then for some stroke patients, right, especially the more severe ones.

00:36:15 Cedric

So just remember that the main purpose of treatment is to get them out of bed and into their functional activities again. So that basically means like we always want the patient to walk on their own again,

because that means if they can walk on their own, they can live happily, right? So in when we're treating stroke patients.

00:36:35 Cedric

Especially the more severe ones. The difficulty comes with the fact that they'll become maximal assist, so that means sometimes when we're getting the that patient to try to walk.

00:36:46 Cedric

For the most severe cases, you'll need 2 or maybe 3 physios just to help assist and bring the patient up and slowly lift.

00:36:54 Cedric

To get them to manually walk, so the two to three physios handling that one patient takes a lot of strain and burden on the like the healthcares, because we already don't have a lot of physios. But if you have to dedicate three of them for like almost like half an hour to an hour and take care of that patient to help manually get them to walk.

00:37:16 Cedric

Then one is they can't do it all day and 2, there's a lot more stroke patients that require the same amount of assistance, but we don't have that many manpower. So definitely is just.

00:37:28 Cedric

The lack of manpower.

00:37:29 Cedric

Is really difficult for stroke patients.

00:37:32 Sebastian

Hmm, I see.

00:37:33 Sebastian

Then so.

00:37:36 Sebastian

Are you familiar with, uh, like some equipment for stroke patient rehab like, like actual names for those equipment?

00:37:46 Cedric

I do. OK. So obviously, you know you, you guys are aware of the robot assistant gate training. That one is the very popular, but another very popular one is.

00:37:55 Cedric

Hold on, let me think. It's the non weight bearing system. I forgot what it's called. Let me see. I can see it.

00:38:02 Cedric

Let me search for you stroke.

00:38:57 Cedric

Yeah. OK, so this this is like by far the most popular one is really just treadmill body weight support system. It's the body weight supported treadmill training.

00:39:08 Cedric

This one is everyone's every rehab's favorite, because the difficulty is they they cannot even support their body weight, right? But you want them to walk. So this one just lifts them in the air.

00:39:20 Cedric

And then you get them back into gait training. So they start moving their legs on the treadmill. So this helps a lot. This is like one of

the best for their treatment.

00:39:29 Cedric

I don't know this I everything else is pretty much manual for machine wise there's the robot assisted gait training which you have already.

00:39:38 Cedric

I I can't see. Hold on.

00:39:42 Cedric

Yeah, this one.

00:39:46 Cedric

This is also very popular.

00:39:48 Cedric

Other than these two, if you're talking about like machine related stroke training, no, there's just basically these two, unless I'm not been made aware of new interventions. But I learned in these two everything else is pretty manually done by the physios.

00:40:06 Cedric

Yeah.

00:40:08 Nadia

Actually, I'm curious. Do you feel like the innovations and physiotherapy for stroke patients like all of these, uh, maybe robot assisted gait?

00:40:19 Nadia

Training.

00:40:20 Nadia

Do you feel?

00:40:20 Nadia

Like these innovations are lagging behind your needs.

00:40:25 Cedric

Lagging behind.

00:40:30 Cedric

I mean, I'd be hard pressed to say yes.

00:40:36 Cedric

But.

00:40:37 Cedric

I wouldn't say I think with the robot assisted gate training already.

00:40:42 Cedric

Well, hard to say, because when you talk about like stroke intervention like our needs like it's really a very like broad base of like what we need. And I can't even comprehend like how some robots can assist.

00:40:57 Cedric

Because I'm not, I'm not in your industry, right? So I don't know, like where you guys can intervene right now, so.

00:41:03 Nadia

In terms of the science of the device, like are we using the right sciences? For example, I think this is based on neuroplasticity, right? We just want to keep the muscle moving so that it.

00:41:14 Nadia

Remembers the moves.

00:41:16 Nadia

Are there like?

00:41:18 Nadia

Uh, maybe it's difficult to answer now because it's a huge question. Could there be other areas in which, like we're not really touching on?

00:41:28 Nadia

Other principles probably, yeah.

00:41:30 Cedric

Other principles.

00:41:36 Cedric

Anything.

00:41:38 Cedric

Let me look at all parts of it.

00:41:46 Cedric

I think right now the only thing I can think of where.

00:41:49 Cedric

Like machine intervention isn't in necessarily right now is like anything related to the sensory motor.

00:41:58 Cedric

Like for neuroplasticity? Yes that that one is right for great gait training, strengthtraining all that. But there's also part in which proprioception and sense sensation, which is a big missing part.

00:42:11 Cedric

Like for example like stroke patients, all of them come up with neglect, which just means.

00:42:14 Cedric

That.

00:42:15 Cedric

On the side of their stroke that they seem to ignore all sensations from that side, but I don't think there's any machine intervention in that part.

00:42:25 Nadia

OK, OK. We can think about it.

00:42:25 Cedric

But yeah.

00:42:28 Nadia

Yeah. Thank you.

00:42:31 Sebastian

OK, so I think this is the last one for the question. So again, this mostly addressed to the people like work with like with the physios, is there like a way that our preferred way of speaking to elderly stroke patients because you know some of them are a bit hard to talk to especially if.

00:42:52 Sebastian

They are not really familiar with physio like concepts like how you guys communicate.

00:42:58 Sebastian

Like effectively like the thing that they need to do to be able to recover. Or is it? Is it easy for those elderly like show patients to pick up what you guys are talking about? Is there like a communication gap, you know, but what what exactly is the problem with communication?

00:43:18 Cedric

I think the biggest part.

00:43:21 Cedric

I think most physios are trained like especially in Singapore. We have quite a strong emphasis on communication.

00:43:26 Cedric

So I don't think that there's much difficulty because especially the stroke patients usually they're quite cooperative, but the biggest difficulty with like communicating between therapists and stroke patient I would say is just the understanding their struggle like just basic empathy.

00:43:47 Cedric

Because it's really difficult.

00:43:50 Cedric

To like establishing a prognosis for the patient because every patient recovers differently. And it all depends on like their premorbid status like how affect how healthy they were and how much that they can recover. So I think a big difficulty is ensuring that you communicate.

00:44:11 Cedric

With the patient like what they want.

00:44:14 Cedric

From the treatment.

00:44:16 Cedric

Because a big difficulty previously, before any communication training was that a lot of physios would try to get them to train the patient to a level from their own goals.

00:44:27 Cedric

So this the the problem was that the patients weren't motivated to reach the same goals. So then there's a lot of frustration along those lines.

00:44:35 Cedric

So.

00:44:37 Cedric

I think other than that, they're.

00:44:38 Cedric

Usually very cooperative, but definitely they would be you have to you have to be extremely patient with them because definitely their mood is a lot worse because they're always tired and it's very difficult to just do basic basic tasks. And there's also the area along like self-confidence because some of them.

00:44:56 Cedric

They also face like urinary incontinence.

00:44:59 Cedric

Stuff like that. So it's just like a loss of basic autonomy. So it's just you have to be extremely respectful in that regard.

00:45:07 Cedric

And the worst part is saying, uh, it's it'll be OK.

00:45:13 Cedric

Because it's very patronizing.

00:45:16 Cedric

But honestly, I yeah, yeah. But I don't know if this is answering your question, but the the main difficulty in communication is just understanding empathy. Just be empathetic. You should be fine.

00:45:27 Sebastian

Alright, lucky, this is actually very useful. Thank you.

00:45:30 Cedric

Oh wait, but that's not an empathy standard if you're talking about, like from a pure healthcare standard, like from them genuinely understanding you, some of them may actually have difficulty understanding your words.

00:45:42 Cedric

Because of their condition, I I forgot about the term is, but like they literally cannot under. Well, I forgot what it's called. It's not aphasia, but it's.

00:45:54 Cedric

Wernicke’s

00:45:56 Cedric

Wernicke’s Aphasia, you see.

00:46:02 Cedric

Or it's the other one brokca’s. No, it's wernicke’s. So it's basically

like they literally cannot understand what you're saying. So like you'll say one thing, yeah, medically, this is a medical problem. So like, because of their stroke now they cannot understand. So that one is the most difficult part.

00:46:19 Cedric

So I'm not sure if there's any robotic intervention there, like if you want to stimulate the uh, we're Wernicke’s Aphasia.

00:46:25 Cedric

So it's literally a part of the brain in the temporal, though, where it like.

00:46:29 Cedric

Processes language.

00:46:32 Cedric

Like so, the difficulty is that since it's damaged because of the stroke.

00:46:36 Cedric

Any words they could hear the words. But like when the words come into the brain like it becomes jumbled up. So like if you say like, do you want the water they may?

00:46:44 Cedric

Hear something like uh.

00:46:46 Cedric

Do you feel like the water is blue or they may not even understand anything like it sounds like mumbling. It's like a different language.

00:46:53 Nadia

OK.

00:46:54 Nadia

How do you deal with that?

00:46:56 Nadia

Sorry. So how do you deal with that as a physio like without robot intervention just typically..

00:47:04 Cedric

One **** of a question.

00:47:06 Cedric

Usually this on hard to answer because this is like uh, this one is known from a physio standpoint. This one is we'll get different discipline. This one is speech language therapy. So that one they'll handle, but if I'm.

00:47:20 Cedric

Also repetition, so I'm not sure if they can read. I'm pretty sure they can still read because I think that's processed differently, so maybe it's like.

00:47:28 Cedric

You just keep repeating the words to help rewire the brain a bit.

00:47:33 Nadia

OK.

00:47:33 Cedric

Yeah. And then speaking slower and then?

00:47:38 Cedric

I heard singing also helps. I'm not sure about that.

00:47:41 Sebastian

Oh yeah, singing does help because it activates a different part of the brain. I read about that.

00:47:46 Cedric

OK, fantastic. Yeah.

00:47:47 Sebastian

OK, so I think in that sense since there is like, uh, I, I have no idea how to say that particular medical term. I'm probably gonna.

00:47:55 Sebastian

Butcher it.

00:47:56 Cedric

Hmm.

00:47:57 Sebastian

Would you say that maybe if instead of like saying it you verbally we can do it with via gestures like umm like for example like visually like looking at a screen like you know how emojis are able to encapsulate certain information just by looking at.

00:48:12 Sebastian

Right. Would you say it would be easier if like I can just present the information visually on the screen for them to understand.

00:48:23 Cedric

I think I think still. Yeah, I think that could help. Like emojis. Yes, but for. Did you mention like writing like any words?

00:48:32 Cedric

I'm not sure if they can read read sorry. I wanna go back on that. I don't think they can read.

00:48:37 Sebastian

No, it's. It's OK. It's OK.

00:48:39 Cedric

Yeah, but emojis I think should be OK. I mean like, like I think displays.

00:48:48 Rizqi

We can just show them video.

00:48:48 Cedric

Sure.

00:48:51 Rizqi

What kind of treatment that we want to do, like show the patient, would that be OK to get them the idea? OK, we cannot do this. We can do this?

00:48:55 Nadia

Yeah.

00:49:00 Cedric

I think that.

00:49:00 Rizqi

Can help? Yeah. OK.

00:49:02 Sebastian

Alright, OK. Nice. OK. I don't have anything else that's actually the last question for me. Do you have anything?

00:49:08 Nadia

I had a.

00:49:08 Nadia

Question, sorry.

00:49:09 Cedric

Oh, yeah, yeah, yeah.

00:49:11 Nadia

There's just one thing that I was curious about is sometimes I hear.

00:49:16 Nadia

That the patients will try to do the exercise, that they'll try to activate, maybe like compensate for with a different set of muscles. Like they'll lift up their hips or something is.

00:49:26 Nadia

That something you just.

00:49:27 Nadia

Try to watch out for?

00:49:29 Cedric

Uh, yeah, that's it. That's a good question.

00:49:33 Nadia

Are there methods that you might use to prevent that?

00:49:36 Cedric

Uh, yeah, that's a good question. So it is like we have two things actually. One is obviously just the the physio themself, we'll make sure like if you're if I want to target the specific muscle, I'm going to make sure you do the exercise correctly. I don't want you to recruit different muscles.

00:49:52 Rizqi

Oh.

00:49:52 Cedric

But typically, I mean, if you're talking like functional exercises like walk in, then as long as the movement looks is correct, then there's no problem with a bit of compensation through other muscles. But of course you're still.

00:50:05 Cedric

Help assist them in the way as long as it's not damaging to their treatment. But if you're talking about like other like forms, we do have like these things like ortho orthosis. I don't know if I'm pronouncing that right. It's like, uh, orthosis? Orthose?. Yeah. So it's like we restrict certain parts to help focus on.

00:50:19 Nadia

OK.

00:50:25 Cedric

The muscles on the right muscles and sometimes.

00:50:29 Cedric

Um, we also just do splits.

00:50:32 Cedric

And the strap. So if I'm not wrong like for example like if we want to increase like thumb muscle activity we split their.

00:50:42 Cedric

Their thumb at the the joint nearer to the wrist. You know, like the.

00:50:46 Nadia

Oh, OK.

00:50:47 Cedric

The big part? Then I'll help activate the thumb muscles. More. Same thing with the the glutes that you were mentioning. We also tape it. We can also tape it on their injured, like their affected side and then they'll help increase glute activation as well.

00:51:02 Cedric

Yeah.

00:51:03 Nadia

OK, OK, I see.

00:51:04 Cedric

Yes.

00:51:06 Cedric

We also take the non-paretic like the non-affected side. We also like restrict that so they don't they don't overcompensate with that side. Yeah.

00:51:14 Nadia

Alright.

00:51:14 Sebastian

Alright.

00:51:17 Sebastian

Hey, umm, do you guys have anything else? I really have nothing.

00:51:18 Cedric

OK.

00:51:20 Rizqi

I think one more one last question from me in terms of.

00:51:27 Rizqi

Device the device design right? Is there any like preference or?

00:51:34 Rizqi

A preference from a medical standpoint like.

00:51:38 Rizqi

Does it has to be?

00:51:39 Rizqi

The how do you say I? I'm not sure what's what. What's the term?

00:51:46 Rizqi

Maybe.

00:51:47 Rizqi

Uh, does it has any standards that has to to meet in terms of the design itself, like maybe a medical standard or the OR like does it has to be from certain material and stuff?

00:52:01 Cedric

The material I-, that one is really sorry, I'm not taught that I'll be honest and that's a that's a different course, but.

00:52:09 Cedric

If you're talking about that, obviously I think the basics are like, just make sure there's a fail safe mechanism, right? Like that one, definitely. And another consideration is the adjustments to the range of motion, because every patient has a different.

00:52:25 Cedric

Amount.

00:52:26 Cedric

So like, especially for the elderly, their range of motion is very poor.

00:52:31 Cedric

So you can, you shouldn't take them if you're doing it passively. Especially you have to be extremely safe because you don't want to overdo it, and if you cause injury then you're just liable for a lawsuit.

00:52:42 Rizqi

Oh shoot. OK.

00:52:43 Cedric

Yeah. So I think it's just about the optimization of like the range.

00:52:48 Rizqi

OK.

00:52:48 Cedric

And the speed also at which the movement is performed, it cannot be fast at all.

00:52:54 Cedric

Because that's not. It's completely not safe, because when you go fast, there's, uh, the muscles will always try to resist due to the some reflex, I forgot the name. Sorry but.

00:53:03 Cedric

So you're supposed to do it slowly.

00:53:05 Cedric

So other than that, I think not really, just slow and fail. Fail safe, make sure it's slow and make sure the range can be controlled.

00:53:15 Rizqi

OK. Yeah. Fair and safe mechanism. OK, that one is important. Yeah, I think that's a problem.

00:53:21 Nadia

Just thought of another question, maybe I could.

00:53:25 Nadia

You talked about some of the measures you used to track progress, but maybe we can also talk about if there's any data that you collect, uh, for patient recovery?

00:53:35 Nadia

Uh. Then maybe we could also try to get with sensors, like, Sebastian and I maybe like, for example, how much force the patient can apply in a certain direction.

00:53:46 Nadia

Etc. Yeah, that kind of data.

00:53:50 Cedric

With regards to.

00:53:51 Cedric

That like, if you're talking about.

00:53:54 Cedric

Does it have to be recorded in the machine or like just in general like the detail looks like?

00:53:58 Nadia

Well, just in general be OK then we can see how it goes.

00:54:00 Cedric

Oh, OK. So let me start off with the the force. You can produce that one is not so important to us.

00:54:08 Cedric

So it is something we consider. But like I said I said earlier, it's just the MMT. So we don't care about the.

00:54:16 Cedric

We don't care about, like, really quantifying it to like the exact like Newton, like force of measure, but as long as they can produce a significant amount of force for their daily needs, it's OK with us. And as long as there's improvement. But the main outcome measures that we look at are mostly functional tasks. So we look at stuff like.

00:54:35 Cedric

We do a walking test, so we just.

00:54:37 Cedric

Get them to.

00:54:39 Cedric

There's a few ones, so there's one for endurance, so that one will just get them to walk for about 6 minutes if they're too tired, maybe 2 minutes or 4 minutes also work. So we just want to get them through 6 minutes and we measure their total.

00:54:52 Cedric

Distance covered and.

00:54:54 Cedric

Of course, for every age group there's a a benchmark.

00:54:59 Cedric

So if this is within reason, then that's good prognosis for us. Another one is gait speed. So like how fast that they walk.

00:55:08 Cedric

Before we can discharge them because there's a specific Community ambulance speed. So that's 0.74 meters per second. So that was the bare minimum limited community ambulance for elderly to be to walk around in the community.

00:55:25 Cedric

Then we also have.

00:55:28 Cedric

Other exercises, like the five times sit to stand or the 32 second to sit to stand, which is basically.

00:55:33 Cedric

How many times or how fast they can sit up and down from a chair, so it's just that one is just, uh, general like strength to see whether that they can stay safe, you know, like stand up from.

00:55:43 Cedric

The chair safely.

00:55:45 Cedric

And I think lastly, not not nothing lastly, but the last very significant one I can think off the top of my head is like a a fall risk.

00:55:54 Cedric

Management so that one is like anything to do with like a single leg stance. So like how long they can balance on one leg or there's this thing called timed up and go, which is how fast they can get up from a chair, dash a bit like walking distance, turn around, walk back and sit back down. Then if they can complete it within a certain time.

00:56:15 Cedric

Then we can suggest that there are we.

00:56:18 Cedric

Can.

00:56:19 Cedric

Quantify like what level of fall risk that they're at then, aside from these measures, every hospital has their own standard of measures, but typically those are the main ones that are followed.

00:56:31 Nadia

Alright, got it.

00:56:35 Sebastian

OK, it's all- Is everyone good? Yep. OK, alright, I got nothing else from my side. So thank you so much. Ced for you know your important contributions because you're you're actually 99% of.

00:56:53 Sebastian

Actually our product because we need to have something to base on. So it's really, really important to have you with us. Thanks. Thanks a lot, man. Thanks. Yeah, thank you. Yeah. So you guys can just leave now. You. Thanks for everything guys. I'll send like inside the group like the information like the recording because I have it.

[END]

Part 2

Nadia 3:34

Okay, so thanks for agreeing to the interview. Before we start, maybe you can run through some of the introduction of our project again, since it’s been quite a while since our last meeting. Okay, so our project is on developing a robotics-assisted rehabilitation device for lower limb rehabilitation for stroke patients. We’re looking specifically at elderly patients, and we’re doing supine exercises. Whether it’s supine or seated, we will probably hammer out fully after we do our site build ups and everything. But for this interview, you can assume it’s going to be mostly supine. The aim of our device will be to support the rehabilitation process by improving patient outcomes and also reduce the physical labor that physiotherapists and other rehabilitation specialists have to do during therapy sessions. For this interview, we are looking at building a use case for the device: where it will be used, how it will be used. The aim is to get a better picture of how the physiotherapist will do the setting up, the actual session itself, and then the storage of the device so that the whole

process is covered.

Nadia 5:01

Okay. So this Zoom meeting is going to be recorded, and we'll be using the transcript for our project. Is that okay with you? So with that, we'll just jump straight into it. The first question that we're looking at is going to be about the frequency, the length of physiotherapy sessions.

UGA 5:28

Okay, so if I just get right into it, frequency would be, usually for acute stroke patients, they'll be inpatient. So this one will be, if I'm not wrong, almost every day of the week, as much as possible. For rehab, it depends. It's usually about an hour, maybe up to 90 minutes at most, because they get quite fatigued doing any more than that. The reason why we want high frequency is because for stroke patients, we just want repetitive practice and more reps so that they can regain most of their function within the first, I think, about three months. That's why we want as many sessions as possible.

Nadia 6:23

Just like, taper off after that? Like you do less sessions after three months?

UGA 6:29

We only do less sessions because usually when there's an acute stroke, there's damage to the brain. In the first about three months, the neuroplasticity of the brain is very high. So that's why in the first three months, we want as many reps as possible for rewiring of functional movement patterns. That's why in the first three months they actually see rapid progress. Afterwards, it's just not as meaningful to do as frequently. Firstly, they already recover most of their function. Secondly, it's not a strategic use of manpower. In healthcare, there's a priority list, so we focus more on acute patients. They'll still receive treatment, just not as much, because we don't expect as many results from high frequency rehab beyond that window.

Nadia 7:40

Okay, yeah. Then for an exercise itself, like let's say knee flexion or knee extension. One thing we noticed is that not every patient is going to do the active movement at the same speed or duration, like each rep. Is that something the physiotherapist determines, or do they consult the patient, like "Can you go further?" or "Can you go faster?"

UGA 8:15

This is patient to patient. The speed, honestly, changes because sometimes, for stroke patients, due to brain damage, they may develop spasticity. Their muscles become stiffer, with increased muscle tone. If they do it too fast, there could be a catch in the muscle. So sometimes they do it slower to reduce that catch. Other times it's slower just because their control of the limbs is poor, coordination is poor. It's not necessarily that the muscles are weak, but there's a lack of connection from brain to muscle. That's why they actively process how to use that limb again, which can make them slower.

Nadia 10:12

We're asking because it'll determine if we make our exercises fixed speed or not. So maybe some way to give the patient control over their own speed.

Rizqi 10:28

Is there like a rule of thumb to determine it? Or a standard procedure?

UGA 10:39

There's no standard speed. We just want the patient to initiate the movement on their own as many times as possible. Honestly, for your device, instead of a set speed, it should be more flexible according to patient needs. For example, if the patient requires assistance, the machine could change the force provided. Most physios just want continuous repetition, and if the patient fails to complete the movement, the machine could help assist to keep the rhythm going.

Nadia 12:01

Okay, you're saying as many times as possible. But is there a limit? Like after a certain number, you say "Okay, you can do this pretty well"?

UGA 12:17

There's no set limit. We don't train non-functional movements; we train functional ones depending on their goals. For example, picking up a water bottle at home. The goal is to repeat that task as many times as possible. In the past, physios could only get 40-60 reps per session manually. With robotics, patients can do up to 1000 reps in one session, and studies showed massive improvement in recovery.

Nadia 14:37

Okay, where I'm getting is that we don't have to aim for a specific number. We can leave it for the physiotherapist to configure.

UGA 14:46

Yes. It depends on motivation, fatigue levels, and is better left for the physio to decide.

Nadia 15:00

Alright, that's great. Sets and reps will be for physiotherapists to configure. Frequency will be about one hour to 90 minutes. Next, treatment plan: we know most recovery is in the first three months. Is there any breakdown within those three months?

UGA 16:07

It's from the start, as much as possible. Once doctors clear the medical issues, and comorbidities are managed, the best exercise is just repetitions. Maybe the first week is medication management and recovery from surgery, but even then they already start mobilizing. Even in week one, if possible, they try to get the patient walking with assistance.

Nadia 17:22

Then following the three months, do any patients come back for more physiotherapy, but at a slower frequency?

UGA 17:37

The three months is just a guideline. Sometimes it extends up to six months, depending on the patient. After discharge, they may come back for regular visits, maybe weekly, monthly, or even twice a year. It's very patient dependent.

Nadia 18:51

We were just asking to get an idea of device usage. Makes sense. Next question: pre-logistics. Where is equipment usually stored?

UGA 19:31

I can't really answer this one. It differs from hospital to hospital. I can ask my profs and get back to you.

Nadia 19:50

Okay, sure. We'll also do a site visit. Next, elderly patients — after stroke, are they lying down all the time?

UGA 20:11

They sit up a lot. They don't lie down all day. If able, they walk. Otherwise, they're in high sitting or reclined positions. It's important for general health, especially respiratory function. If they lie down all day, their respiratory function declines.

Rizqi 22:11

So for rehab sessions, it's recommended to have them seated instead of lying down?

UGA 22:20

Yes. Not lying flat all day.

Nadia 22:39

Next: use cases for our device, called Kinetics. Besides hospitals, where else could it be useful?

UGA 23:18

Most important is in community hospitals. Stroke patients go there immediately. In hospices, not so much. Hospice is more palliative, so

robotics might not be used much for exercise, though maybe for small assisted movements. But mainly community hospitals.

Nadia 25:37

Okay, so most relevant in community hospitals.

UGA 25:41

Yes. I haven't really heard of robotics being used in acute hospital settings.

Nadia 25:53

Next is technical. About sensor data we could provide physiotherapists:

- Range of motion / joint angles
- Angular velocity
- Smoothness of movement
- Force / strength estimation
- Repetition counting / segmentation / trajectory
- Muscle fatigue
- Consistency of movement

What do you think?

UGA 26:50

All are useful. For fatigue, we usually use RPE (rate of perceived exertion) or daily fatigue scales. Angular velocity is used less, since we focus on functional data like gait speed. Strength is very useful, good correlation with outcomes. Range of motion is also very important, since we know minimum required angles for functional movements. Everything is useful, except angular velocity which is more for research.

Nadia 28:28

Got it. We'll emphasize ROM and strength. Next, disinfection. Our device needs to be cleaned. Doctors sometimes use disposable sheets, or sprays, or alcohol wipes. Which is preferred?

UGA 29:41

I haven’t really heard of disposable sheets except pillow cases. Usually beds have sheets swapped out regularly. For devices, they use alcohol wipes, standard hospital grade. Comfort is also important — especially for elderly, because their skin can be fragile.

Nadia 28:28

Super nitty gritty, yeah, I see. Okay, great. Then when we present it, we’ll focus more on ROM and strength. Sounds like those two are most important. Alright, great. Next: disinfection. Our device will need to be cleaned at some point. We wanted to check — doctors sometimes use sheets over the bed, like disposable paper? Would that be preferred to something like a spray or alcohol sanitizer?

UGA 29:12

Paper on the bed?

Nadia 29:14

Not sure what to call it. Rizqi, do you know? Like disposable sheets you swap out between patients when they’re lying on a bed.

Rizqi 29:29

Even bed sheets, right?

Nadia 29:31

I don’t think so. Maybe thinner disposable ones.

Rizqi 29:35

Is it disposable?

Nadia 29:38

Yeah, they swap it out. I’m not too sure.

UGA 29:41

No worries. I haven’t really heard of disposable ones except pillow cases. Usually the bed just has its own sheet, and the nurses swap it

out regularly. For sanitization, they use alcohol wipes. Hospital standard ones, I forgot the brand, but it's the same everywhere. You can even buy it from hospital suppliers. They just use those wipes for devices, like gait trainers, wheelchairs, everything.

Nadia 30:36

Okay, noted.

UGA 30:42

One more thing. If you're targeting elderly patients, comfort has to be a priority. Not just for user satisfaction, but because their skin is extremely fragile, paper thin. If the device moves roughly, it can cause skin tears very easily. That leads to infection risks, and in community settings patients often already have comorbidities. Many are on medications like Heparin, which is a blood thinner, so if their skin tears, they won't stop bleeding. Especially stroke patients, they'll almost always be on blood thinners. So comfort isn't optional, it's essential for safety.

Nadia 32:09

So whatever touches the skin must be soft, cushioned.

UGA 32:16

Yes, straps should be gentle, maybe padded with foam. You want to avoid chafing or pressure sores.

Nadia 32:22

Okay, avoid chafing. Got it. Next question is on setup and storage of the device, but you mentioned earlier you haven't seen much of that.

UGA 32:35

Yeah, I haven't seen storage setups. For setup, with electrical or robotic devices, we try to make it fast. Sessions are short, 60 minutes on average. So setup should take no more than 10 minutes. For example, with EXO (the exoskeleton), the first time they measure femur length, tibia length, pelvis width, and so on. After that, it only takes about five minutes to strap it on. Usually patients are seated in a chair in

a walking area — sometimes just a long hallway. Then they strap the ankles, below the knee, and mid-femur, and tighten around the pelvis. Once secured, the patient stands and starts moving.

Nadia 34:44

Okay, that's the setup process. Do patients ever complain? Anything we should watch out for?

UGA 34:55

Not really. Complaints are usually just minor — nothing about the device itself. The only recurring one is some patients, both elderly and younger, don't like using robots because it makes them feel less independent. That's the main psychological barrier.

Nadia 35:42

Okay, that's useful to know.

Nadia 35:46

Wait, I had a question… can't remember it. Never mind. Last question: before starting a session, what kind of preparatory questions do you ask? Like joint pain? Sleep schedule okay? Mental state?

UGA 36:26

Aside from the basics like pain, most of it is rapport building. If we ask structured questions, it's usually part of a standardized assessment. For example, the Functional Independence Measure (FIM) — we ask how well they can perform daily tasks, like showering or using the bathroom, whether they need assistance.

Nadia 37:14

Would that be tracked session by session?

UGA 37:21

No, not every session. More like every two weeks to a month, depending on where the patient is in recovery. In the first three months, maybe more frequent. Later, maybe only twice a year. If independence is already established, the measure becomes less important — unless the

patient reports a decline, then we reintroduce it.

Nadia 39:04

Okay, got it. We can work with that. We won’t keep you any longer. Thanks for having us.

UGA 39:10

Alright, thank you.

Monica (Occupational)

Interviewee Consent and Profile

The following questions are to build a profile of the interviewee in their relevance to the development of the device, as well as confirm their experience with stroke patients and/or movement specialisation.

Name	OTA
Occupation (Physiotherapist, Kinesiologist, or Physiatrists, etc.)	Occupational Therapist
Date	09/09/25
Our Introduction	We are a group of final-year students from the National University of Singapore (NUS), currently working on a research project as part of the CDE4301 Final Year Project module. Our project focuses on the viability and development of a robot-assisted rehabilitation device targeted at stroke patients' unilateral lower limb/extremity (LE) to be used in supine or sitting exercises. Our device aims to support the rehabilitation process by improving patient outcomes while also reducing the physical workload on physiotherapists and occupational therapists during therapy sessions. We are currently gathering professional insights to guide the design, especially the user needs, usability, and practical relevance of our solution.
Scope of Interview (Comes first)	Thank you for agreeing to participate in this interview. The purpose of this interview is to gather expert insights that will inform the development of a robot-assisted rehabilitation device for the lower limb aimed at supporting stroke patient recovery. What your participation involves:

	• A short, recorded, and noted, interview lasting approximately 30-40 minutes. • Questions related to your professional experience with stroke patients, particularly in movement rehabilitation. • No personal or identifying information about patients will be requested. Your rights: • Your participation is completely voluntary. • You may decline to answer any question or stop the interview at any time. • Your identity will be made anonymous if requested. The information you provide will be used solely for academic or research purposes.	
Consent Statement	I understand the purpose of this interview and consent to participate.	Y/N
	I agree that the information I provide can be used anonymously for research and documentation related to this project.	Y/N
	I agree to be recorded for transcription purposes.	Y/N
Verification of Patients' Experience	What is your experience with stroke patients in terms of years or number of patients?	What settings have you worked in (acute, rehab, or outpatient)?
	Varied in terms of age, recent increase of 40 year olds in the in-patient side, the first stroke used to be more ~50, but currently age presentation has decreased to 40, majority not quite elderly	Currently covering the neural unit in a hospital setting, acute and sub-acute wards, before woodlands was in Stroke Support Station, S3, has experience with subacute

	20 (rare and not lifestyle-related) - 80 year old stroke patient range. Mostly, 40 year old stroke patients.	towards chronic.

Questions (To be organised)

Question	Rationale (What this question can help with)	Response
Patient Evaluation		
What is the ability of patient in diff stages		Acute (1 week post-stroke): Medical stability critical; BP fluctuations; fatigue, drowsiness, alertness vary. Sub-acute: Medically stable; therapy intensity increases; mobility assessed; focus on standing, weight shifting, gait training. Chronic: Device could be used in cycles across multiple patients with PT supervising.
Comments of SGH video		1. Likely caregiver education video (bedside, non-functional exercises). 2. Could be passive stretches to do at home. 3. Bed exercises prescribed case-by-case (e.g.,

		3×/day, 10 reps; or warm-up 5 reps before standing). 4. Generally only a small part of the sub-acute; used if the patient is very weak.
Weight range for device to be placed on bed	**ME** Make sure the weight is easy to carry	5kg is doable, but the process of putting it up and the space needed, patients have sheets, cushions, etc on the bed, must be considered,
Device needs		Requires high level of customization from patient to patient for factors such as leg length, number of movement planes, automized changes in movement type.
Device setting up	**ME** Make sure the device is easy to set up (later do a setting up testing)	The device has to be easy to set up. Easy to clean and slides
What is your typical initial physical assessment of stroke patients, and what are the criteria you use to determine treatment readiness?	**UI/EE/ME** Understand our demographic	OT (Occupational Therapy) do not usually do lower limb assessment, geared towards upper limb usually Physicians do balance assessment, however.

What measures do you use to track progress?	**UI/EE/ME** For deciding which data to collect/sensors to use	First measure: mobility in daily activities (walking, shifting, dressing). More dependent patients → more equipment and training. National scales used: Barthel Index, EQ-5D (quality of life, mood, pain, subjective health).
Comments on our UI		Kahoot-like screen might be beneficial for elderly patients. They may have difficulty logging into the app. Gamification strongly encouraged (patients bored, increases motivation) Calendar function not very useful inside health facilities—better if physiotherapists set up exercises. Need a good external cue that moves with the patient, make it stop when it hits a target etc

		Can focus less on patient log in, and make it something that physiotherapists set up for patients to use ***CONSIDER -> having a resting page that can just be left open for patients to tap and check their scores and upcoming appointments, not calendar
Customization of exercise for patient		Patients have varying ability levels for different exercises and strength level is not even throughout Movement A of exercise 1 is easy, but movement B of same exercise is difficult.
What are some ever used mechanical or robotic assistance during physiotherapy on the lower limb (LE)? Please elaborate on the types of devices you have employed: (Try to get their purposes and the challenges faced)	**UI/EE/ME** When is it that physiotherapists choose to employ devices so that we know what pain points we are addressing	Devices used in physio mainly depend on patient dependency level and need for additional support. Pain points: bulky devices, storage, setup time, extra workload for therapists.
What kind of exercises are performed on the LE in supine and sitting position?	**UI/EE/ME** For us to understand better about the degree of motion of	The SGH video is a correct example of bedside exercises Hip movement

	our device	Supine and sitting bedside exercises: hip and ankle movements critical. Ankle movement is very important and often missed. Bed exercises are best for very weak patients.
Compensatory movements can hinder recovery. What are some common LE compensatory movements that occur and what do you do to minimize them? (Using the hips/momentum instead of executing controlled movements)	**UI/EE/ME** Planning how our device can minimise 'wrong' motions so that it is effective, or how the UI can give warnings	Patients may use hips/momentum instead of controlled movement. Devices/UI should minimize "wrong" motions or provide corrective feedback.
How do you typically communicate recovery to patients? OR what is the feedback that patients need from you during a session?	**UI** How should the UI communicate progress after each session?	Having an incentive system can motivate the patients. Recovery communicated through progress in daily activities and mobility. Scores or feedback in-app can help with motivation.
What are some difficulties you face when conducting physiotherapy sessions? (Large number of patients, physically taxing, etc)	**UI/EE/ME** Pain points again	Large number of patients, physically taxing for physiotherapists. Equipment must not increase workload.
How many stroke patients do you usually see a day?		Large caseloads in hospital setting.

What is the frequency of sessions? What is the length of sessions, and how do you determine the duration of one rep of exercise?		Session frequency depends on stage of recovery; reps and session length tailored to patient stamina.
What goes into the design of a treatment plan?		Based on mobility, medical stability, readiness, and goals (independence, ADLs).
Are elderly patients lying down in bed during rehab sessions? Or are they seated?	**ME** To determine the device design (mechanism)	Often seated; lying in bed only if very weak or acute.
What are your thoughts on the feasibility of KINETIX in different contexts? (E.g. Hospice settings, etc.)		Patient abilities vary within the same exercise: one movement may be easy, another difficult. Exercises should adapt to differing strengths across motions within the same exercise.
These are the metrics that we aim to communicate to physiotherapists to monitor progressive recovery. What are your thoughts on their usefulness? Range of Motion (ROM), Joint Angles Angular Velocity & Smoothness Force / Strength Estimation (Normalised to body weight) Repetition Counting & Segmentation Trajectory Tracking Muscle Fatigue Estimation Consistency of Movement		ROM and strength useful. Smoothness and fatigue important for rehab progression. Scores as motivators are very useful.

Transcript
9 September 2025, 04:17am

Disclaimer: This interview transcript has been lightly edited for brevity and clarity. Filler words, repetitions, and unrelated exchanges have been removed, but the meaning, tone, and conversational style have been preserved.

Ron 0:04
OK, so maybe let me give you some context first. So actually we are final year students from NUS working on one of the final project modules CD4301. So our project focuses on the viability and development of a robotics-assisted rehabilitation device targeting the lower limb, either in supine or sitting exercises, so actually we aim to support the rehabilitation process by improving patient outcomes, while also reducing the physical workload of physiotherapist and occupational therapist.

So we currently are gathering professional insights to guide the design, especially for the user needs which are the physical therapies and occupational therapies. We also want to see if our device usability and the practical relevance of our solution, so maybe I want to start with asking actually what is your experience with stroke patients in terms of their age, number of stroke patients that you have seen?

OT A (Institution A) 1:05
Hmm. So right now I am in one of the teams in the hospital where one of our where we cover the neural unit. So both from the acute phase to the sub acute phase. So right after post row and then as well as as they go through the inpatient rehab. So we do have our subacute wards as well. Before being in Institution A, I was in the community. So I
was in the stroke support station. We had focused on charity. I don't know if you have heard of them. Then, I also see patients after stroke but that is a much more subacute towards chronic face like post.

OT A (Institution A) 2:04
Age-wise I would say very, very I see as young as 2030 year olds that have strokes and also older in their 80s that have the stroke. So each wise is quite varied. Recently we have a lot more 40 year olds after stroke in the inpatient cycle.

OT A (Institution A) 2:28
Yeah. So years, I guess I also don't know how to count. Sorry. Maybe like if you think, yeah.

Ron 2:28
OK.
No.

Sebastian 2:30
Yeah.
I do.

Ron 2:35
Would there be a larger proportion in like the older age range or actually they are quite spread out?

OT A (Institution A) 2:43
Oh, OK. Oh, OK, OK, I would say. Actually on the first stroke right? Usually we have a. There used to be a lot of 50 year olds, 50-60 year olds. By the time we are talking about when it's that, let's say it's a stroke onset in the 80s. There is also, but actually a lot of that like fifty 60s. But in our hospital. Now we see a lot of 40 year olds.

Ron 3:12
OK.

OT A (Institution A) 3:12
Yeah. So there's this. I would say it's not true that the majority of them are elderly. We actually see a lot like that middle age range.

Yeah, of course. The really young population, the 20s and 30s, that's a lot rarer, like usually they may have a stroke, not so may not be lifestyle. They have a bleed. Yeah, that is the 20 year olds, yeah.

Ron 3:39
I see. OK. OK. So yeah, they are so more common now that stroke patients are like 40 years old and a kind of middle age range right here. It's popping up more often.

OT A (Institution A) 3:44
Yeah. I guess it's not really just like it's not really that now, but all along there is this trend that's about 50-60 then after that. But recently in this population, I can't speak for the rest of the country, but in this area, right, we have a lot of 40 year olds that have their first stroke, yeah.

Sebastian 4:17
Sorry, I just have a question to ask. So how do you usually track the progress of stroke patients regarding let's say mobility, especially when it comes to walking on the lower limb aspect?

OT A (Institution A) 4:21
OK, so as an OT, we don't really do the lower limb assessment per SE, but our physios will do. The BBS, the balance scale, is the one.So the ministry has this one rehab scheme. I'm not sure if in your research you have come up on it. So it is actually how the ministry funds the rehab services that the medical, the therapy and then medical therapy we have. Down from upstream to downstream, so one rehab starts from the acute episode in the hospital and then to the sub acute in patients we have and then the downstream is like your community we had like for example S3 is your community rehab. So there's this one rehab programme where the ministry is providing subsidies.

One of the outcome measures they track is the Berg Balance Scale (BBS), which assesses mobility and balance. Another is the Modified Barthel

Index (MBI), which evaluates overall function in activities of daily living (ADLs). OTs are more involved in the MBI, since it covers not just mobility but also components such as dressing, showering, and feeding. Generally, performance on the BBS and the MBI is correlated. Patients who score better on the BBS tend to perform better on the MBI. For example, being able to stand more steadily usually means the patient can also dress more independently.

The One Rehab scheme also tracks the EQ-5D quality of life scale, which assesses patients' perceived health. This is a brief questionnaire that asks about mood, pain, and overall subjective health, producing a score from 0 to 100. It provides an additional patient-reported outcome measure.

OTs also use upper limb assessments, but those may not be directly relevant here.

Sebastian 7:24
Thank you.

OT A (Institution A) 7:26
Oh sorry, there's also a Frenchay they have. They are going to change it, but I don't know, I can't remember which is the one they changed to. But basically we also track an instrumental IADL scale.

Just now I mentioned Barthel Index right, which is a self care basic activities of daily living. So like showering, feeding, and getting dressed. Then once they go into the community or outpatient, the other thing we want to track is the instrumental activities of daily living. So things like going back to hobbies, going back to work, taking public transport.

That is the IADL assessment. We used to use this assessment called the Frenchay IADL scale, but I think now they are changing to a different IADL scale.

I think mobility ties in a lot with instrumental ADLs. Definitely when mobility increases, a lot of patients are then able to go back to all these IADL.

Sebastian 8:30
OK, alright, thank you.

Ron 8:32
Yeah.
Prof Idris 8:32
Gee, OT A, I'm just curious between acute to sub acute to the chronic, we have in terms of the ability of the patient what are the differences?

OT A (Institution A) 8:39
OK, so of course at the start when a patient is in the very acute phase of stroke, there are the stroke symptoms, like the weakness that is present. But there's also this element of medical stability. For example, their blood pressure generally fluctuates at that acute phase after the stroke. They might have drowsiness, fluctuating alertness and fatigue levels. Yeah, that is more apparent in the acute phase.

Prof Idris 9:14
It's when emergency emergency services will be provided right in that sense.

OT A (Institution A) 9:29
So for example, right after the stroke, in that first one week, the body is still adjusting to the blood pressure change. So we will see that the blood pressure still fluctuates. Doctors will be more lenient in what they allow the blood pressure to fluctuate at.

You will observe patients that are more fatigued, they fluctuate in and out of arousal. If the stroke is very severe, the arousal level will fluctuate. Fatigue levels also, they will be more easily fatigued.

Then after that acute phase, into the sub acute phase, when they are medically more stable, their blood pressure starts to stabilise. That’s when therapy can pick up in intensity.

Prof Idris 10:14
OK, so in this period, sorry, in this period the therapy that we are referring to. So let's say if for instance of like the lower limb, what what kind of therapy will usually be so recommended?

OT A (Institution A) 10:24
It goes with the performance of the patient. Of course the first thing is always we assess — the physios, the OTs — we will assess the mobility first. It really depends on the patient. For very severe strokes, even getting them to sit up and transfer out of bed, getting onto a chair, or getting onto a chair for showering, that is already mobility training for them.

Then getting them to practise standing, standing balance, weight shifting towards their affected side — that is also already progress in their mobility. Step by step, when they can do that, then the next is actual movement, like the physio will start gait training.

For gait training, physios use different aids. Patients who are more dependent will require different forms of walking therapy, aids and equipment.

Yeah, so that’s how the mobility training progresses, if that’s what you’re asking.

Prof Idris 11:36

So I think here we are curious to know, the team found a video from SGH, about eight years old. In this video, the therapist is working with a patient lying down on the bed, doing leg exercises. Maybe someone can show the video here for PTA to have a look. OT A, I think we are very curious. So for this and this, this is meant for bedridden patients, right?

OT A (Institution A) 14:21
Yeah, so this depends. If patients are not ready and I'm assuming this video is a caregiver education video to show caregivers how to assist then these can be bedside exercises for someone who just had a stroke in the hospital, if they are not yet ready for things like sitting up, standing, or walking.

At the same time, it could also be stretching, because I can see it's a passive exercise. It can also be a stretching exercise for caregivers to do when the person goes home. So it really depends on the intention. But these are just passive ranging.

Prof Idris 15:29
Would this have to be done like on the regular interval once a day or twice a day?

OT A (Institution A) 15:33
Regular interval meaning, usually if the physio prescribes these exercises, they will specify to the family, like maybe three times a day, 10 reps each movement. But it depends on the person. Sometimes they may even say these exercises are just to warm up before the person does other types of exercise, like standing or walking at home. If it is a warm up, then it will not be so long, maybe 5 reps of each movement.

Prof Idris 16:16
So OT A, in your opinion, this is what this exercise with something that you see in the sub acute setting?

OT A (Institution A) 16:25
I will say that it's a very small part of therapy. Usually if the physios are doing these passive exercises, it's more of a warm up and then they will move on to other exercises. Of course, if patients are very very weak, then yes, they may be doing these passive exercises and after that trying to actively move on the bed on their own.

Prof Idris 16:51
OK, so there is actually an advantage to have. Maybe before we go there, because I think it's quite hard work to lift the leg and do the repeated motion. If there is a device that can help with this, or take some load off from the physiotherapy.

OT A (Institution A) 17:11
I think it depends on the feasibility, because nowadays there are devices like lower limb pedal you can use on the bed. They fix it over and then patients can use it on the bed. But the feasibility and practicality is a bit challenging, because you have to pick it up, set it up, connect it every day, bring it away, wipe it down.

At the end of the day, using something like that may increase the load on the therapist rather than reduce. If it’s just me, I go over, move the patient, and leave — it’s faster than having something to set up.

Prof Idris 17:50
OK. But if it's something like what would be like a recommended acceptable sort of weight range for such a device to put on the bed?

OT A (Institution A) 18:12
Uh.
But this is a bit a bit tricky question because—

Prof Idris 18:20

Because a bag of rice is like 10 KG right? So is anything less than 10 KG or less than five KG something that is quite doable?

OT A (Institution A) 18:23
Yeah, I think something not too heavy, like if you mention around 5 KG, it's not too heavy, quite doable. But it's the process of it. How fast is it to put up, how much space you need.

Because usually patients are on their bed, they have a lot of things on the bed. They got maybe three or four wedges and cushions, then their blanket, their sheet, incontinent sheets, their cuff pumps. So actually a lot of things around the patient's bedside.

Prof Idris 18:51
Well, I think if it's just maybe about if it goes into something like a leg brace and then it occupies only sort of the half the bed for the lower limb, yeah.

OT A (Institution A) 19:20
Possible, but possible. Then another thing is also does someone need to supervise when the patient is using it? Like how safe is it if we don’t need to be there when the patient is using it passively?

Prof Idris 19:44
Yeah, so actually we’re also thinking, let’s say sub-acute is still in the hospital ward setting. If the patient is bedridden and then they likely will suffer from bed sores and so on, they might require these exercises.

OT A (Institution A) 20:28
They might be. They will require these exercises, especially patients that are usually in the bed. They're more dependent.

Prof Idris 20:38

So we're thinking actually, whether you think it's possible to have to like in that community care setting, there are few beds, right? And then a few of these patients will require this exercise and if a device is set up for like 3 patients and then they are all started at the same time, the physiotherapist just needs to maintain an overview.

OT A (Institution A) 20:40
Yes.

Prof Idris 20:57
You know it’s going to be a slow warm up cycle, and then just maintain an overview to make sure things are going well and the patients are OK. Would that be something useful, you think?

OT A (Institution A) 21:10
Hmm. I'm thinking.

OT A (Institution A) 21:15
It probably is, but have to see the practical use of it. How easy it is to set up, how easy it is to clean, because of infection control. How easy it is to clean down.

Also how adjustable it is, because different patients their leg length is different. Not just leg length, but also their comfort level. If you are talking about people in the sub acute phase or later phases of stroke, if they have a bit of tightness, then they may not tolerate maximum range or full range. This will differ from patient to patient.

Is it adjustable? That I’m also wondering. I’m also wondering what are the movements that can be done. Is it all the different movements, or only one movement? Because if it can only do one or two movement planes, then is it worth the time setting up?

Versus if I set up this thing and it can do this movement, then that movement, then another movement, like different planes of movement, and

it is automatized so I don’t need to come over and set it again, then maybe it’s easier.

Prof Idris 22:51
Oh, I think the team was thinking of using an app that the official therapist could control.

OT A (Institution A) 22:55
Oh, OK. Oh, OK, that means using the app to change the movement, is it?

Prof Idris 23:04
You can potentially change the movement, yes.

OT A (Institution A) 23:07
Oh, OK, OK.

Nadia 23:08
Yeah.

Prof Idris 23:09
And then I think changing the range is a good idea as well if we can, yeah.

Rizqi 23:13
OK.

OT A (Institution A) 23:15
I think that would set it apart a bit from the things already in the market. In the market got this pedal, that kind of pedal normally patients have to sit up to do the pedalling, either sit on the side of bed or sit out on a chair. But I think there is a pedal bike that can be pushed into the patient's bed, so when lying down they can do.

But since it's a bike, you cannot really adjust. It's just one movement. At most you can adjust how much assist you get and how much

you do yourself. But if your patient is not full range, you cannot use the bike because of the movement. And it also only goes one way, it doesn’t do other things like rotation or ankle. It just does one movement.

The use of that is a bit limited. Still good to have patients moving — always better than not moving — but the actual therapeutic benefit is limited lah because it's only one type of movement.

Prof Idris 24:32
Yeah, so I think maybe the team was thinking about this. Thank you very much, I think those are very important points for them. What they were thinking is that because some of the exercises in the video you saw just now seem to be quite repeatable, they have to do for a certain number of goals. If those exercises take up quite a bit of time and are quite strenuous on the physiotherapists, would it be good if we could replace that with the device?

Prof Idris 25:05
And then having that for a couple of patients. The other thing they were thinking is, let’s say if the patient does regain some mobility but still don’t have balance, so still lying on the bed but got a bit of muscle strength already coming back.

Then if this device can be switched to an active mode, whereby the patient can exert their own force against some resistance, progressive resistance, would that be useful?

OT A (Institution A) 25:32
Oh I think that will be useful. Yeah, having the ability to switch from passive to active assisted mode will be helpful. Another thing that is helpful from the patient’s perspective is anything that motivates them.

So is there anything on this device that can be motivating? Like today I do this, you know, some of the exercises. But now they have this

visual, so they say, oh you finished one round while sitting down, you finish one round around the track. Then sometimes got a goal like, oh I did this and I get a star at the end. Like there’s a target.

We do notice in the ward, patients are very competitive. Once you bring in something with a score, like oh you did this much, you get 150 points, another patient will say, I can beat that, I will do 200 points. So it’s universal. This competitiveness, this motivation.

Prof Idris 26:44
mm, I think if there's some autonomy also over it is nice. Yeah. So I think what Nadia was thinking of is the app, not just for the physiotherapists but also for the patient to log in. They can view their progress, they can view how well they are doing.

OT A (Institution A) 27:00
Oh but if they have to log in, right, they must know how to use it and also need to have a phone. The thing is, in the ward, a bit difficult lah. Some elderly don’t have their own phone.

Prof Idris 27:31
So what would they like? Like stickers?

OT A (Institution A) 27:38
So for example, some of the upper limb robotics, they will have a screen like that. They have a computer with the robotic exercise, so the patient will do the exercise, and on the screen there will be something like a game that tracks how many they did. Then it will trigger like, oh, you did this amount today.

Prof Idris 28:01
OK, a display.

OT A (Institution A) 28:08

Then some of the bikes, of course it's the bike with the screen in front. I don't know whether this will come with a tablet or something, but then they can see it. They can see like, oh, I am doing this. Then when they are finishing, it can also help them to calm down. If not, they will feel very tired just lying there.

Nadia 28:14
Yeah. Yeah.

OT A (Institution A) 28:28
Especially if it's active assist mode right, then they just lie there doing, doing, doing, not sure when it's going to end. Then it becomes quite stressful for patients if they are just doing and don’t know when the exercise will finish.

Prof Idris 28:31
Yeah. No. Yes.

OT A (Institution A) 28:45
Yeah, yeah, yeah.

Prof Idris 28:45
Yeah, good point, good point. Because initially we were thinking of an app on the phone. But what you said makes a lot of sense, especially if they are very old and they don't have phones or they are not so able to log into the app and all that. I think we should consider that. Maybe let's go back to the team, you all have slides right, that you want to show.

Ron 29:21
So maybe this is the design that we are looking at right now. This is the overall structure of the preliminary design we have. It supports the kind of passive exercises that we saw in the SGH video.

Not only passive exercises, but also active exercises. This device can do knee extension, where the elderly or patient can fully extend their leg. It can also do lateral movements like hip abduction.

What we are trying to achieve with this device is to allow patients to reach a certain range of motion for their joints, and also build on their muscle strength through the active mode.

Sebastian 30:21
And then continuing on from Ron. As mentioned previously, all of the data being used here is sensed by sensors already installed on the device. These readings are then sent to the tablet, or an app on your phone. The app contains the patient's information, like how much force they are exerting on the device, whether it is enough, whether they have been improving.

What Nadia, my fellow electrical engineer undergraduate, did is make it able to track progress and predict how much to tweak the difficulty level of the device. For instance, if after 5 sessions you've been improving really well, say you go through three different levels of difficulty, by the 6th session it will increase it for you. This makes it easier for the physio, nurses, or OTs, since the device can automate the process and adjust the exercise on its own.

To help out, I'll give a brief description of how the app works. As you can see here, this is the profile page of the person using the device. From the patient point of view, you can see different stages on the top right. Stage 1 is the easiest, Stage 4 is more difficult. The "affected level" just means which leg is being used — left or right.

It also tracks personal information like age, weight, and so on. As you move on further, it also keeps track of appointments. For example, if an elderly patient needs to go to the community hospital weekly for exercises, they can keep track of it on their phone.

OT A (Institution A) 32:31
OK.

Sebastian 32:44
So as you can see, we can store all the appointments here for the physio as well as the patient to keep track of when they need to go for their rehabilitation appointments.

From the physio side, they can pick and choose what kind of exercise they want to use and they are also able to decide and modify how difficult it is. For example, if they regress in terms of performance and there's another relapse of stroke, then they will need to revert it back to active assistance. Basically the device will move the leg for them, without the elderly or patient exerting their own force because there's been a relapse.

The device also actively tracks how much force the person is using throughout the session. This allows the physio to understand whether the user is able to exert force on the foot rest, if something is wrong, or if there is a particular part of the movement where the user cannot continue. That's how it is supposed to track for our device.

If problems happen, the patient can just press the assistance needed button so the physio or nurse can come over to help in that situation.

Prof Idris 34:47
Perhaps we get some feedback from OT A. OT A, what do you think about the interface?

OT A (Institution A) 35:04
OK, I just wonder how much the patients will use the calendar function. This is one thing I'm thinking — if this device is used in the ward and the therapists are the ones setting it up for the patient, then what is the calendar going to be used for?

If the patient is in the outpatient setting, this device is only looking at a very specific few exercises and movements. It’s not in that sense the most big part of therapy. So when I see the app with a calendar, that part feels a bit disconnected from the rest of the device.

I do really like the part where you have a very visual image. It’s a very visual cue. In neuro we say this is a very good external cue to tell patients when they are putting in the movement. And I think your thing blinks, right, like it moves? So that’s an external cue we use a lot in neuro rehab, because it moves along with the patient and gives that cue as they are doing the exercise.

Prof Idris 36:21
Yeah.

Nadia 36:21
Yeah.

Sebastian 36:23
Yeah.

Ron 36:25
Yeah.

OT A (Institution A) 36:34
It’s a very good concept to have. It’s just about how to make it goal oriented. Like if it’s just going like that, right, is it going to be like going bigger, bigger, bigger, then hitting the target for example. Then that will be a true external cue. When they hit that target, they know that it’s their maximum. that target. Then they know that it's their maximum.

OT A (Institution A) 36:55

That comes back to small again, then go go go, keep the target. Yeah. I just wonder, ’cause I look at the thing you’re designing right, it’s basically you got the idea from the knee CPM machine, correct.

The knee CPM is just passive movement right? When you say you want to make it active assisted, which part is both? Is it the flexion, the hip flexion assisted, and then the knee extension assisted, you thinking like that?

Rizqi 37:17
OK.

Sebastian 37:30
OK. Yeah. So for the active mode, sorry. You wanna take or I'll take?

Ron 37:30
Aye. Oh, no. Yeah, you can do it.

OT A (Institution A) 37:36
Mm-hmm.

Sebastian 37:37
OK. So essentially how the mode works is that for passive essentially the device does the work for the patient so no force exerted on the patient. Now for the active one, it's a bit more different in a sense because what we're doing is that we have sensors on the foot rest on different parts of the foot rest.

OT A (Institution A) 37:45
Yeah, yeah.

Ron 37:45
Mm-hmm.

Sebastian 37:55

So let’s say for example they’re trying to do horizontal movement back and forth. What we’re going to do is set the threshold on the sensor. If the patient does not go above that threshold, then basically the machine won’t move.

They need to maintain that force on the foot rest to keep it moving. Same goes for the other movements also.

OT A (Institution A) 38:17
Yeah, yeah. So is the movement this way. It is the hip and knee extension that is assisted, not the flexion. The hip and knee flexion will always be passive because of the way it is designed. Right. And then the heat and knee extension is the one that is assisted.

Sebastian 38:38
No, there's resistance throughout the whole movement, so both pushing, pulling and the sideways as well.

Ron 38:46
So should we also have an active mode for the knee flexion and hip flexion like when the patient pulls back, oh, OK, OK.

OT A (Institution A) 38:50
Oh yeah, so that’s what I was wondering, correct. I’m wondering how, when you actually use it right, because when we see patients, their strength — if this target is for neuro patients, it’s a bit different.

For ortho patients, actually this knee CPM you see, the ones using this machine the most are the ortho units, like post TKR. They put them on the knee CPM machine, and really the purpose is passive movement of the leg, to help with joint movement.

But if you talk about neuro patients, it’s different. Sometimes their weakness is very weak. They may be able to bring up their leg, but they have a lot of weakness in extension. Similarly, they have very strong

adduction, but very weak abduction. Because it’s neurological, the movement is not the same — the recovery is not even, not equal throughout.

So depending on the machine design, if the resistance is the same for extension and flexion, certain patients may not be able to use it effectively. They will struggle. They may have a lot of strength in one movement, but struggle in the other.

Ron 40:36
OK. So yeah, so our device is to cater to both knee, knee flexion and knee extension as well, right. OK.

OT A (Institution A) 40:36
Hmm. That'll be ideal, but I don't know how you will put the sensor. Yeah, yeah, yeah then.

Ron 40:46
Thank you. Right. OK. Other than that, our device also wants to do hip abduction and hip adduction exercises, which currently this kind of CPM devices cannot really do. So I think that is one improvement we want to look at. Would that be useful?

OT A (Institution A) 41:12
Actually I will say the hip movements are useful, but I think ankle movement is very useful. Because ankle tightening happens very quickly. For ankle movement, usually it takes the longest time for them to regain, compared to the bigger muscles.

So ankle movement is something that is quite missed out. There are many things out there to help them passively move their ankle, but not so much beyond that.

Prof Idris 41:56

No, I think it's possible because they are thinking of the pedal and having four sensors on it. So it might be possible to have a movable pedal, in the sense of a ball and movable. Yeah, OK.

Prof Idris 42:27
OK, then that's fine. But if it does, then OT A, you're on your lunch break right now, is it?

OT A (Institution A) 42:35
Yes, at 1:00. It's my lunch.

Prof Idris 42:36
Oh, dear. Then you're not going to have lunch for today.

OT A (Institution A) 42:39
Oh, no, no, no. My lunch is at 1:00.

Prof Idris 42:42
01 o'clock, OK. Then I think because the guys are going to S3 on Thursday, but I suppose you are at Woodlands Health right now. What does the team say? Do you have more questions for PTA, things you want to close the loop on? Or you want to maybe take it in two stages — they work on the prototype first, then when midway done can show it to you. If that's OK with you, PTA, for your advice.

OT A (Institution A) 43:10
Sure, sure, sure, sure. You're not going down to S3. I am meeting Shimmy now.

Ron 43:16
Benjamin.

Sebastian We're meet, yeah.
43:17

OT A (Institution A) 43:17
Oh, Benjamin. Oh, OK. OK, OK. I think he will give you a lot more insight. Like as a physio.

Ron 43:18
Yeah.
Yeah.

Sebastian 43:24
OK.

Ron 43:25
OK.
OT A (Institution A) 43:25
Yeah, yeah, yeah. I think he'll be able to advise a lot more.

Prof Idris 43:32
But OT A, what you have said just now is very, very useful. At least I think the team knows that. Oh, they are very sure about where they are, you know, situating the problem and why is it that they have to work on? Yeah.

OT A (Institution A) 43:32
Oh, sorry.
No worries. Hope it has been helpful and have fun at S3.

Ron 43:49
Yeah. Thank you so much.

Prof Idris 43:56
Are you OK with, like, maybe if they still have some, you know, small questions. I suppose Ron has your contact in it.

Nadia 43:57
Yeah.

OT A (Institution A) 44:04
Good. Ron has my e-mail. Yeah, yeah.

Ron 44:08
Yeah, yeah.

Prof Idris 44:10
So you prefer questions through e-mail. Is that?

OT A (Institution A) 44:14
No, no problem actually. Are you able to teams me? I'm not sure 'cause you.

Prof Idris 44:20
Just teams, loh.

OT A (Institution A) 44:22
I don't know. As in by chat, I don't know whether you can teams me or not.

Ron 44:26
Can, ah.

Prof Idris 44:27
I think Ron, you can set our, we can set our teams chat and then add OT A in. I'm not sure whether but you have to log into because you'll be invited as a guest.

Ron 44:28
I'm not sure actually.

OT A (Institution A) 44:42
Oh oh.

Prof Idris 44:42
But we can try that and then we might be able to team CES.

OT A (Institution A) 44:45
Oh, OK, OK. OK. Otherwise, yeah, yeah, I'm OK. If you e-mail me or if you want to just have a quick if you got any question, you call it. We have a quick chat over teams. Like let's say like a quick call for teams is also fine. Yeah, can just let me know that.

Prof Idris 45:06
OK, very, very nice of you. Thank you so much.

OT A (Institution A) 45:09
No, thank you. Thank you all the best.

Ron 45:10
Right. Thank you so much, OT A, thanks for your time and supporting us also thank you.

Sebastian 45:12
Thank you. Thank you so much. Thank you for your time.

Nadia 45:12
Thank you. Thank you.

OT A (Institution A) 45:15
No, no problem. Thank you. Thank you. Thank you. Bye. Bye. Bye bye.

Prof Idris 45:17
Thank you and.

Ron 45:18
OK, have a good day. Thank you so much. Bye.

Ben (SSE)

Interviewee Consent and Profile

The following questions are to build a profile of the interviewee in their relevance to the development of the device, as well as confirm their experience with stroke patients and/or movement specialisation.

Name	PT A
Occupation (Physiotherapist, Kinesiologist, or Physiatrists, etc.)	
Date	05/07/25
Our Introduction	We are a group of final-year students from the National University of Singapore (NUS), currently working on a research project as part of the CDE4301 Final Year Project module. Our project focuses on the viability and development of a robot-assisted rehabilitation device targeted at stroke patients' unilateral lower limb/extremity (LE) to be used in supine or sitting exercises. The aim of our device is to support the rehabilitation process by improving patient outcomes while also reducing the physical workload on physiotherapists during therapy sessions. We are currently gathering professional insights to guide the design, usability, and practical relevance of our solution.
Scope of Interview	Thank you for agreeing to participate in this interview. The purpose of this interview is to gather expert insights that will inform the development of a robot-assisted rehabilitation device for the lower limb aimed at supporting stroke patient recovery. What your participation involves: ● A short, recorded, and noted, interview lasting approximately

	30-40 minutes. ● Questions related to your professional experience with stroke patients, particularly in movement rehabilitation. ● No personal or identifying information about patients will be requested. Your rights: ● Your participation is completely voluntary. ● You may decline to answer any question or stop the interview at any time. ● Your identity will be made anonymous if requested. The information you provide will be used solely for academic or research purposes.	
Consent Statement	I understand the purpose of this interview and consent to participate.	Y/N
	I agree that the information I provide can be used anonymously for research and documentation related to this project.	Y/N
	I agree to be recorded for transcription purposes.	Y/N
Verification of Patients Experience	What is your experience with stroke patients in terms of years or number of patients?	What settings have you worked in (acute, rehab, or outpatient)?

Questions (To be organised)

Question	Rationale (What this question can help with)	Response
Patient Evaluation		
What is your typical initial physical assessment of stroke patients, and what are the criteria you use to determine treatment readiness?	**UI/EE/ME** Understand our demographic	Physiotherapists check if patients can move first, then try activities such as rolling over, getting out of bed, or walking. They gauge the level of assistance required. Some patients can't even roll over; progression depends on independence level. Warm-up exercises are often prescribed before walking. Patients can come as early as 2 weeks post-stroke or many years later.
What measures do you use to track progress?	**UI/EE/ME** For deciding which data to collect/sensors to use	Progress is checked with level of assistance (e.g., how much support is needed), gait kinematics (e.g., leg swing, step length). For sub-acute recovery (2 weeks–2-3 months stay), gait and functional ability are key indicators. Anything quantifiable and objective is useful.

What are some ever used mechanical or robotic assistance during physiotherapy on the lower limb (LE)? Please elaborate on the types of devices you have employed: (Try to get their purposes and the challenges faced)	**UI/EE/ME** When is it that physiotherapists choose to employ devices so that we know what pain points we are addressing	Examples include: Exoskeletons (Ekso Bionics) requiring leg and hip measurements, runs on batteries; setup time should ideally be <5 min. Anti-gravity treadmill with shorts for unloading; CPM machines (but evidence shows limited value since too passive). Whole-body vibration machines for balance and muscle activation. Tube tables for bedridden patients (expensive). Availability often depends on rehab centre and cost.
What kind of exercises are performed on the LE in supine and sitting position?	**UI/EE/ME** For us to understand better about the degree of motion of our device	Supine/seated is usually manually assisted and more labour intensive. CPM used but not effective long-term. Cycling/pedalling devices are common, but patients struggle to generate strength independently. Flexion/extension exercises are typical; less equipment exists for hip adduction/abduction. Adjustability to patient

		strength and lower limb dimensions is important. Supine/sitting better for acute patients. Portability and orientation flexibility (supine, seated, standing) is ideal.
Compensatory movements can hinder recovery. What are some common LE compensatory movements that occur and what do you do to minimize them? (Using the hips/momentum instead of executing controlled movements)	**UI/EE/ME** Planning how our device can minimise 'wrong' motions so that it is effective, or how the UI can give warnings	Legs tend to fall out sideways → safety harness used for gait training. Compensations include using hips/momentum instead of controlled movements.
How do you typically communicate recovery to patients? OR what is the feedback that patients need from you during a session?	**UI** How should the UI communicate progress after each session?	Feedback is usually about level of assistance and independence. Some machines have cameras showing kinematics, foot placement, step length. Patients need to see tangible improvement in ability to complete tasks (not just gamification). Gamification works for a while but patients eventually want real task independence.
What are some difficulties you face when conducting physiotherapy sessions? (Large number of patients, physically taxing, etc)	**UI/EE/ME** Pain points again	Large patient numbers, physically taxing (esp. manual supine/sitting therapy). Setup time for

		devices (goal <5 min). Some devices are heavy (need wheels/trolley). Portability is important in multi-floor hospitals, though not S3.
Do you think there is a lack of equipment specialised for stroke patient rehabilitation? And do you believe that this device can assist their rehabilitation? If not, why do you think so?	**UI/EE/ME** Justification for research and development for device	Yes, especially for hip adduction/abduction. Most devices target flexion/extension only. Current equipment availability varies by rehab centre. A customisable, portable, adjustable device that can be used in supine, seated, and standing would address gaps.
Physiotherapy Session		
What is the frequency/length of sessions, and how do you determine the duration of one rep of exercise?		Sessions usually 15–30 mins in hospitals. Number of reps and sets prescribed by therapist based on evidence. Start with minimal proven value and progress upwards.
How to design a treatment plan?		Therapists check clinical evidence, then prescribe progressive exercises. Transition happens when patients regain some function (Joint angle becomes important only after initial recovery)
Where is the equipment stored?		Equipment is usually kept outside for shared use. Larger devices often come with trolleys (e.g., CPM). Portability is key in bigger hospitals.

Are elderly patients lying down in bed during rehab sessions? Or are they seated?	**ME** To determine the device design (mechanism)	Both supine and seated therapy are used, especially in acute and bedridden patients. Seated and standing preferred when possible, as lying down is less valuable long-term.
What are your thoughts on the feasibility of KINETIX in different contexts? (E.g. Hospice settings, etc.)		Could be valuable if adjustable and portable. Needs to target a wide range of patients (young teens → >100 y/o). Hospitals already have some equipment; community/hospice may need more portable solutions. Sensor data is also valuable as physiotherapists want quantifiable data as far as possible to track patient progress.
These are the metrics that we aim to communicate to physiotherapists to monitor progressive recovery. What are your thoughts on their usefulness? Range of Motion (ROM), Joint Angles Angular Velocity & Smoothness Force / Strength Estimation (Normalised to body weight) Repetition Counting & Segmentation Trajectory Tracking Muscle Fatigue Estimation Consistency of Movement		All are useful if quantifiable & objective, some concerns about how muscle fatigue will be estimated.
Which method of disinfection is preferred by physiotherapists (disposable paper or sanitizer, etc)?		Disposable wipes/paper or spray sanitiser. Surfaces must be wipeable & scratch-resistant.

When using larger tools/devices for physiotherapy, what is the setup and storage process like?		Typically outside storage; wheeled in when needed. Setup time must be <5min ideally. Heavier tools need wheels/trolley.
Before starting a session, what kind of preparatory questions do you ask? For example: 1. Any joint pain? 2. Have you gotten enough sleep recently? 3. Mental state?		Physiotherapists usually check for joint pain, sleep quality, and mental state. They also ask about comfort with the device (fit, straps, positioning) before starting. During the session, they frequently check if the speed, resistance, or intensity feels alright for the patient.

Other Notes:

Legs tend to fall out sideways, safety harness used for gait training, distance and speed, ui interface is more informative and lacks vr and targeted, assistants usually help with exercise, pt does prescription. There are shorts to help with anti gravity treadmill, there is a camera in the machine to show positin of feet and check kinematics of each leg, step length etc

Progress is checked with level of assistence typixally, gait kinenatics if legs are swinging antueally or if they are, these is for sub acute revover, between 2 weeks to 2-3 minths stay

Can come to s3 as early as 2 weeks post stroke or many years later, but patients are very varied in terms of level of independence ie cannot stand up etc, hospitals also have rheir own rehab so may not come here,

Goal is to increase level of idnependence, whole body vibration machine that is for balance, perception, high frequency is muscle activation, equipment is typically outside to be used, exoskeleton requires physio to take measurements of the patients leg and hip to adjust the exoskeleton, runs on batteries -> ekso bionics, want to minimise set up time to 5min.

Supine seated and gait training

Supine and seated is typically more manually involved, cpm evidence wise so far is not useful because it is very passive focussed. Lot of pedalling on bicycle, it’s difficult for them to generate

strength on their own, device must be able to adjust according to patient strength, try to have the device be used in different orientations to maximise usefulness

Bedridden patients, tube table that loads the feet, normally quite expensive, most of beds are height adjustable, we are starting to see younger patients in their teens and 100s, 50/ 60s more common that 30s40s, availability of tools depends on cost and physio, machine shld target a wide grp of people and be customizable

Materials that should be chosen are those that can be wiped down easily and scratch-resistant,

In hospitals, 15

OTHER NOTES

The therapist typically prescribes number of sets etc, they will check evidence and then start at the minimal number with proven value.

Warm up exercises prior to walking —> prescribed by therapist

Portability is important, therapists prefer them to be up and moving, just lying down in not most valuable, this is better for the acute patients, adjustability of weight is important, electrical stimulation can be considered

How to decide when to shift over to a different exercise, normally physiological therapists will just get them to move first and try some activities such as getting out of the bed, walking, and the PT will gauge the amount of assistance required for patients to perform the task. Some patients can't roll over in the bed etc.

A machine like this will help.

Anything quantifiable and objective is useful for the physiotherapist.

Metrics are useful, gamification is fun but some patients may not even be able to tell where the screen is etc, but only works for a while until they wanna move on to being able to complete the task

Not much for hip adduction and hip and hip abduction, availability of equipment is dependent on the rehab centre

- Our device is going to be a part of the session, but not the whole session. Both seated, standing, and supine would be ideal (similar to press machines in hospitals, but limited to flexion and extension)
- Footplate make swivel able
- Device can be both laid on bed frame and placed on the floor
- For stroke, joint angle is important only after some level of function is regained
- Try to get clients
- Power plugs are not too much of an issue
- If heavy need wheels, cpm machines usually come with a trolley, manual carrying 5kg max, some physics cover many floors

Executive Summary

COMBINED SUMMARY OF PRELIMINARY DEVICE FEEDBACK		
Pain Points	**Positives**	**Action Items**
ELECTRICAL		
No fixed standard for assistance force, therapists just “help enough” (Interview 1)	Active-assisted mode more valuable than fully passive (Interview 1)	Implement adjustable assistance levels instead of fixed torque.
Need strong safety measures (emergency stop, insulation, power cut) (Interview 2)	Open to electrical devices if safe and easy to disinfect (Interview 3)	Prioritise fail-safes, simple plug-ins, and clear indicators.
Setup time must be minimal (<10 min) (Interview 2)		Minimize need for device configuration
MECHANICAL		
Bulky/heavy devices hard to handle/store (Interview 2)	Weight around 5kg is acceptable for bed devices (Interview 2)	Design lightweight (<5–7kg) modular parts.
Patient discomfort risk: skin, chafing, infection (Interview 3)	Passive still valuable for preventing contractures (Interview 1)	Use soft, wipeable materials with adjustable straps.
Cluttered bedsides, limited space (Interview 2)	Adjustable supports useful for different patients (Interview 3)	Prioritise compact size, compatible with bedside clutter.
	Seated/inclined setups are common (Interview 1)	
SENSING		
Exact force quantification less useful clinically compared to normalised values (Interview 1)	ROM, strength, fatigue, smoothness are meaningful (Interview 2)	• Focus on functional metrics (ROM, reps, fatigue, smoothness).
Patients often compensate with hips/momentum (Interview 2)	Visual feedback helps reduce compensation (Interview 1)	Consider which metrics can detect compensatory hip movements.
USER INTERFACE		
Elderly struggle with complex UI/logins (Interview 2)	Gamification motivates (scores, rewards, competitiveness) (Interview 2)	Instead of personal device-oriented, build tablet-oriented therapist-driven setup; patients only see progress display
Calendar/logging features not useful in clinic (Interview 3)	Simple progress dashboard useful (Interview 3)	Replace calendar with progress/goals dashboard.
Difficult communication with patients, need empathy (Interview 1)	Visual/audio cues effective (Interview 1)	Support a variety of visual/audio/simple feedback modes.
Motivation drops if difficulty not adapted (Interview 1)	Adaptive difficulty boosts engagement (Interview 1)	Implement adaptive difficulty.